Transformer-Enhanced Multi-Agent Reinforcement Learning for Adaptive Honeypot Deception

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Declaration

It is hereby declared that

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3. The thesis does not contain material which has been accepted, or submitted, for any other degree or diploma at a university or other institution.
4. We have acknowledged all main sources of help.

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Abstract

Modern cyber attackers have been increasingly using automated tools to fingerprint and to evade static defensive measures. Deception technologies, such as honeypots are important for proactive defense, but their present static nature which presents the same predetermined vulnerabilities and responses makes them identifiable by sophisticated adversaries. This thesis proposes a novel Transformer-Enhanced Multi-Agent Reinforcement Learning (MARL) framework for the creation of an adaptive honeypot swarm dynamically adjusts its deceptive responses to the behavior of an attacker in real time. The proposed system uses a Transformer encoder to understand attacker motive through sequence analysis paired with a centralized training decentralized execution (CTDE) MARL framework for the coordination of deceptive actions across the swarm. The research addresses important aspects of adaptive cyber defense such as modeling attacker behavior, coordinating multiple agents as well as optimizing dynamic deception strategies. By combining deep learning and reinforcement learning the system makes it possible for the honeypot swarm to learn complex engagement strategies, extend interaction with attackers and maximize information acquisition capabilities. The result plays a part in promoting AI-driven cyber deception technologies and, by demonstrating that sequence modeling when coupled with multi-agent coordination can enhance significantly defensive effectiveness in case of sophisticated cyber threats.

Keywords: Cyber Security, Transformer, Adaptive Honeypot, Multi-agent Reinforcement Learning, Intelligent Honeypot Systems, AI-driven Defense, Sequence Modeling, Self-Supervised Learning, Centralized Training Decentralized Execution (CTDE), MITRE ATT&CK Framework, Intrusion Detection, Threat Intelligence Collection, Honeypot Swarm.

Dedication (Optional)

A dedication is the expression of friendly connection or thanks by the author towards another person. It can occupy one or multiple lines depending on its importance. You can remove this page if you want.

Acknowledgement

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Chapter 1

Introduction

1.1 Background

The cybersecurity domain is undergoing a major shift which are caused by attackers who now use highly sophisticated automated tools to bypass traditional defense mechanisms [1]. Recent cyber threats, especially Advanced Persistent Threats (APTs) are highly adaptable and use multi-stage attack patterns to remain undetected for long periods of time [2]. By 2025, cybercrime is expected to have an annual economic impact of about $10.5 trillion USD which is indicating the critical need for flexible cybersecurity frameworks to keep up with the technological advances of modern attackers [3].

Cyber defense strategies today are heavily dependent on deception technologies such as honeypots [4]. The primary objective of honeypots is to be used as a decoy and make sure that attackers do not come close to the actual resources, at the same time serving as a source of valuable threat intelligence. However, the traditional honeypot has some limitations in itself, as it is static, meaning that it gives predefined vulnerabilities and responses that can be easily detected by experienced attackers [5]. It has been proved that experienced attackers can do "pre-attack checks" on fingerprint honeypots, making them ineffective for intelligence gathering as well as attack diversion [6].

The application of artificial intelligence to the field of cybersecurity is a real step forward in terms of defense. Deep learning models have shown excellent performance in intrusion detection, and Transformers have been shown to be significantly better than traditional recurrent models such as Long Short-Term Memory (LSTM) and Recurrent Neural Networks (RNN) [7, 8]. This is because they utilize self-attention, which makes them excellent at modeling long-range dependencies and complex sequential patterns with properties that make them uniquely suited for analyzing attacker behavior over time [3].

At the same time, Reinforcement Learning (RL) has also been recognized as an effective approach for building adaptive defenses that can adapt to the dynamic threat environment [2]. Recent studies have also validated that Reinforcement Learning (RL) based honeypot deployment strategies are more effective than traditional static rule-based defense mechanisms [4, 9]. Multi-Agent Reinforcement Learning (MARL) further extends this idea by enabling multiple autonomous agents to collaborate in developing defense strategies and addressing the limitations of centralized decision making [10].

Although there have been major breakthroughs in the development of Transformer-based threat detection and Reinforcement Learning-based adaptive defense mechanisms, the integration of these two technologies for coordinated honeypot swarm defense is still a major gap in the current state of the art. The most effective of these systems, such as the framework proposed by Siddiqui et al. [10], leverage MARL for adaptive response but still rely on entropy analysis and fuzzy logic for threat detection, focusing on statistical anomalies rather than reasoning about attacker intent. Similarly, although single-agent adaptive honeypots based on RL have been developed [4, 9], these systems are not capable of network-wide coordination.
This thesis therefore proposes a novel integration of Transformer-based deep attacker behavior analysis with Multi-Agent Reinforcement Learning to enable coordinated honeypot swarm defense. Unlike the existing systems that are based on statistical detection or single-agent optimization, the proposed Transformer-MARL framework will enable honeypots to analyze the semantic context of the entire attack launched by the attacker, which will enable the honeypot swarm to provide strategic and coordinated responses to complex multi-stage attacks. This, therefore, marks a major milestone in the development of intelligent and coordinated cyber deception systems.

1.2 Rationale of the Study

The fundamental limitations of the current honeypot systems and the demonstrated potential of AI-driven approaches to overcome these constraints are the motivation of this research. Regardless of whether they are low-interaction or high-interaction, traditional honeypots are widely recognized for their rigid and predictable behavior, which skilled adversaries can quickly detect and evade. This results in short interaction times and poor intelligence collection [9]. The honeypot's ability to deceive and gather intelligence is seriously undermined by this predictability which leads to "empty logs and missed learning opportunities."
The rapid growth of Internet of Things (IoT) devices has significantly expanded the attack surface, introducing resource-constrained systems that are especially vulnerable to exploitation [11]. These kinds of heterogeneous, dispersed settings demand defensive mechanisms that can function independently and coordinate across assets which are challenges for centralized security systems. The multi-agent system provides a viable alternative by enabling coordinated responses and distributed intelligence without introducing single points of failure.
Furthermore, it is the success in the adaptive single-agent honeypots that demonstrate the untapped potential of the learning-based deception. HoneyIoT [4] has accomplished remarkable results in 467 malware samples being collected vs 14 by conventional systems and having a Shodan Honeyscore of 0.0 (perfect deniability of cover). These findings confirm that RL-based adaptation significantly enhances honeypot effectiveness, further motivating the extension of this approach to multi-agent coordination for network-wide defense.
This research is further motivated by the advancement of Transformer architectures in the cybersecurity domain. Wu et al. [7] demonstrated that Transformer-based intrusion detection systems achieve F1-scores exceeding 99% and substantially outperform LSTM and other traditional models. The FlowTransformer framework [8] provides systematic evidence that Transformers excel at capturing sequential pat-

terns in network data precisely and this capability is needed for understanding complex, multi-stage attack sequences.

1.3 Problem Statement

The current status of cybersecurity defense does not have a system that can dynamically coordinate multiple honeypot agents based on a deep understanding of attacker intent to deceive sophisticated attackers. The existing honeypot systems have several critical drawbacks that this thesis aims to overcome:
First, honeypots using *static deception tactics* are prone to fingerprinting attacks. Sun et al. [6] showed that traditional honeypots can be detected with 92% accuracy using probe packets that target variations in error-handling behavior. This is a critical drawback that makes it easy for attackers to distinguish honeypots from real systems and simply avoid them altogether.
Second, the *absence of multi-agent coordination* makes it challenging to effectively defend against attacks. Although single-agent adaptive honeypots have been promising [4, 9, 12], these honeypots are standalone and lack the ability to coordinate their responses on a network level. This drawback is explicitly acknowledged in the existing literature, where Li et al. [2] state that their single-agent strategy "faces challenges such as limited computing power and data silos" and "struggles to adapt to subsequent attack stages of APT."
Third, honeypots based on *state decision making* using Reinforcement Learning (RL) are incapable of understanding the complete context of attacker campaigns. Existing honeypot systems typically make decisions based on immediate state observations or statistical features, which lack the comprehensive sequential understanding required to detect complex, multi-stage attacks.

1.4 Objectives

The proposed Transformer enhanced Multi-Agent Reinforcement Learning (MARL) Honeypot framework has some specific and well defined objectives which it aims to achieve. Those are as follows:

1. **Development of a Simulated Environment for the Interactions between Honeypots and Attackers:** To create and implement a simulation environment of configurable networks which includes swarms of three to five honeypot agents and a well scripted attacker based on MITRE Adversarial Tactics, Techniques, and Common Knowledge (MITRE ATT&CK) framework for supporting training and evaluation of deception policies based on AI by providing a reproducible testbed.

2. **Design, Implement and Pre-train a Transformer-based Attacker Behaviour Encoder:** To design and implement a Transformer encoder model which is trained on the sequences of the actions of attackers by using self-supervised learning through masked token prediction. This produces context embedding vectors which capture the intent of the attacker and the progression of the attack campaign.

3. **Development and Implementation of a Centralized Training Decentralized Execution (CTDE) Multi-Agent Reinforcement Learning (MARL) Framework:** For developing and implementing a multi-agent system based on established multi-agent reinforcement learning (MARL) algorithms such as multi-agent deep deterministic policy gradient (MADDPG) or Q-value mixing (QMIX) where Transformer context embeddings are incorporated by central critic and dynamic deception actions are executed by each honeypot.

4. **Design and Specification of Action space and Reward function:** To design and specify honeypot actions which includes modification of dynamic service banner, alteration of filesystem structure as well as controlled pacing of data leaks and to define a reward function which encourages sustained attacker engagement along with deeper interaction within deceptive environments.

5. **Conducting Comparative Evaluation Against Established Baseline Methods:** To thoroughly evaluate the proposed framework against static honeypots, state-of-the-art single adaptive honeypot and multi-agent reinforcement learning (MARL) based methods but without the implementation of any sequential models like Transformer using metrics such as depth of interaction, attacker dwell time as well as generalization to novel attack patterns.

1.5 Methodology in Brief

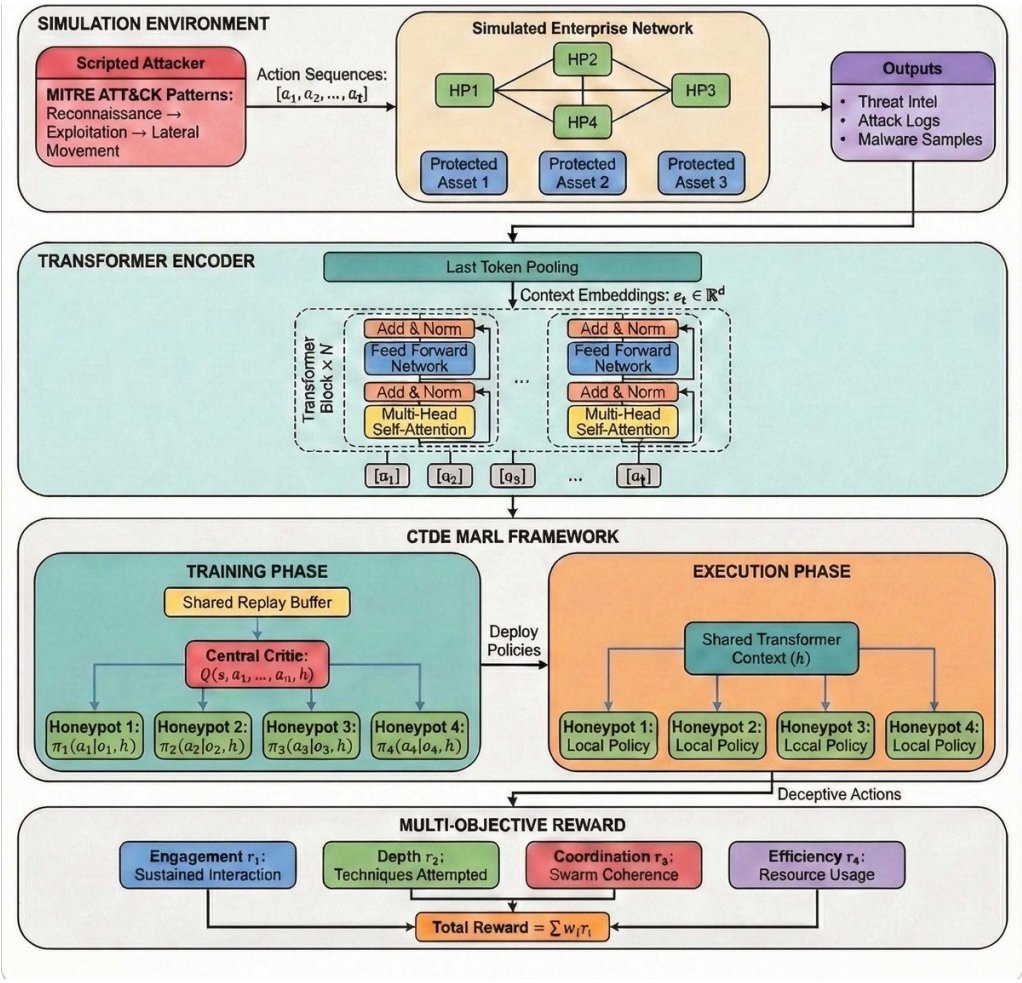


Figure 1.1: Complete architecture of the Transformer-Enhanced MARL Honeypot Framework. The framework consists of four integrated components: (1) **Simulation Environment**—containing a scripted attacker generating action sequences based on MITRE ATT&CK patterns (reconnaissance, exploitation, lateral movement), a simulated enterprise network with five coordinating honeypots (HP1–HP4) protecting critical assets, and an output module collecting threat intelligence, attack logs, and malware samples; (2) **Transformer Encoder**—processing attacker action sequences $[a_1, a_2, ..., a_t]$ through multiple transformer blocks with multi-head self-attention, feed-forward networks, and layer normalization, followed by last token pooling to produce context embeddings $e_t \in \mathbb{R}^d$ capturing attacker intent; (3) **CTDE MARL Framework**—implementing centralized training with a shared replay buffer and central critic $Q(s, a_1, ..., a_n, h)$ that incorporates transformer context, while enabling decentralized execution where each honeypot operates independently using local policies informed by the shared transformer context to generate deceptive actions; and (4) **Multi-Objective Reward**—combining engagement reward r_1 (sustained interaction), depth reward r_2 (techniques attempted), coordination reward r_3 (swarm coherence), and efficiency reward r_4 (resource usage) into a weighted total reward $\sum w_i r_i$ that guides policy optimization.

The proposed solution integrates a parametric pipeline which is based on state-of-the-art computational approaches by integrating advanced deep learning techniques along with multi-agent reinforcement learning. Figure 1.1 exhibits the complete architecture of the proposed Transformer enhanced Multi-Agent Reinforcement Learning (MARL) Honeypot framework which clearly illustrates the integration of all major components and their interactions.

Transformer Encoder for Modeling Attacker Behavior: The sequences of an attacker's actions will be processed by a Transformer architecture to produce contextualized embeddings. According to the findings of FlowTransformer [8], 'Last Token' classification head along with the encoder based architecture will be employed which will provide optimal performance. Self-supervised masked token prediction based on synthetic attack sequence will be used to pre-train models following MITRE Adversarial Tactics, Techniques, and Common Knowledge (MITRE ATT&CK) patterns. As illustrated in Figure 1.1, the action sequences are received by the Transformer Encoder from the synthetic simulation environment and it also produces context embeddings which capture the semantic meaning of attacker behaviors.

Multi-Agent Reinforcement Learning Framework: A centralized training with decentralized execution (CTDE) architecture will be implemented by the system either using multi-agent deep deterministic policy gradient (MADDPG) or Q-value mixing (QMIX) algorithms. A central critic has access to global state information as well as Transformer embeddings of all agents' at the time of training. However, using only its local observations and the shared Transformer context each honeypot agent operates independently at the time of execution. By using this approach Siddiqui et al. [10] achieved 97.24% detection accuracy with similar architectures. The Centralized Training with Decentralized Execution (CTDE) multi-agent reinforcement learning (MARL) Framework block in Figure 1.1 exhibits this approach with the training phase on the left along with the execution phase on the right.

Specification of the Dynamic Deception Action Space: Each honeypot agent can select from actions which include simulating different vulnerability profiles by modifying service banners, altering visible filesystem structures, controlling the pace and content of fake data leaks and mimicking different system loads by adjusting response timing. This design builds upon proven adaptive honeypot implementations which were reported in prior studies [5, 9]. The Simulated Enterprise Network in Figure 1.1 displays the five honeypots (HP1–HP5) that coordinate to execute these deceptive actions.

Reward Function Design and Specification: The multi-objective reward function is designed to encourage interaction time which is the reflection of prolonged attacker engagement, techniques attempted which refers to deeper penetration within the deception system, effective coordination among honeypot agents along with effective utilization of system resources which is in alignment with the design principles proposed by Wushishi et al. [13]. The bottom part of Figure 1.1 shows the multi-objective reward component which illustrates the process of combining four reward terms to form the total reward signal which guides policy learning.

1.6 Scope and Challenges

Scope: The design as well as evaluation of an AI-driven adaptive honeypot swarm system operating within a simulated network environment is the main focus of this research. The scope of this research encompasses:

- Development as well as evaluation of the framework within a simulation environment modeled on attack patterns defined by the MITRE Adversarial Tactics, Techniques, and Common Knowledge (MITRE ATT&CK) framework.
- Support for coordinated operation of a honeypot swarm which consists of three to five agents.
- Emphasis on scripted adversary behaviors that represent common Advanced Persistent Threat (APT) techniques which include reconnaissance, exploitation and lateral movement.
- Evaluation metrics aligned with established cybersecurity benchmarks which include attacker dwell time, interaction depth, and detection evasion rates.

Challenges: A number of significant challenges must be resolved for successful implementation and deployment:
Training Data Scarcity: High-quality labeled data which captures sophisticated attacker behavior sequences is challenging to obtain. In contrast to other fields where data collection is relatively straightforward, cybersecurity research is constrained by ethical and legal issues when collecting real attack data [14].
Algorithmic Challenges in Multi-Agent Coordination: Without any centralized control achieving effective coordination among multiple autonomous agents introduces significant algorithmic challenges, particularly in maintaining policy convergence and avoiding local optima [10].
Computational Overhead: The framework must balance sophisticated reasoning capabilities with real-time response requirements as Transformer models and Multi-Agent Reinforcement Learning (MARL) algorithms are computationally intensive (Wushishi et al. [13] achieved 2.07 ms decision latency).
Adversarial Robustness: The framework must maintain its effectiveness against attackers who attempt to recognize and manipulate defensive mechanisms driven by AI [3].
Generalization: Ensuring that the learned policies can generalize to novel attack patterns not seen during training remains a fundamental challenge in machine learning-based security systems [13].

Chapter 2

Literature Review

2.1 Preliminaries

This section establishes the fundamental concepts and technological foundations relevant to adaptive honeypot systems, multi-agent reinforcement learning, and Transformer-based threat modeling.

With a focus on current developments (2024–2026) in adaptive honeypot technologies, multi-agent reinforcement learning for cybersecurity, and Transformer-based threat detection, this survey of the literature looks at pertinent research papers published between 2021 and 2026. To illustrate the development of integrated intelligent defence systems, papers are arranged strategically.

Honeypot Technologies: Honeypots are security resources intended to be probed, attacked, or compromised. They have two functions: they collect threat intelligence and deflect attackers from legitimate assets [1]. Conventional honeypots are categorised according to the degree of interaction: high-interaction honeypots offer full system settings with better intelligence-gathering capabilities but higher risk, whereas low-interaction honeypots offer restricted services and emulation [4]. Conventional honeypots are fundamentally limited by their static nature, which allows sophisticated opponents to fingerprint and evade them due to predefined replies [6].

Transformer Architecture: Originally developed for natural language processing, the Transformer architecture has shown impressive performance in a variety of sequence modelling problems [7]. Transformers are able to capture intricate patterns and long-range dependencies without the sequential processing limitations of recurrent models thanks to the self-attention mechanism. Transformer-based models have reached cutting-edge intrusion detection performance in cybersecurity, with F1-scores surpassing 99% on standard benchmarks [7, 8].

Reinforcement Learning for Cyber Defence: Through interaction with an environment and rewards that direct behaviour toward desired goals, reinforcement learning (RL) offers a framework for agents to learn optimum policies [2]. RL allows for adaptive responses in cybersecurity applications that can change as threat landscapes do. While algorithms such as proximal policy optimisation (PPO) and deep Q-networks (DQN) offer useful mechanisms for policy learning, the Markov decision process (MDP) formulation captures the sequential decision-making character of defensive measures [4, 13].

Multi-Agent Systems: This technique expands single-agent reinforcement learning to situations with several interacting agents. The centralized training with

decentralized execution (CTDE) paradigm uses global information during training to enable independent execution while addressing the issues of partial observability and coordination [10]. This strategy is especially applicable to distributed defence systems, where honeypots need to cooperate without continuous communication.
MITRE ATT&CK Framework: A thorough understanding of adversary strategies and tactics which are seen in actual cyberattacks is offered by this framework [15]. The framework further supports systematic classification of adversary behaviours across different stages of the attack lifecycle such as reconnaissance, initial access, execution, persistence and lateral movement. It is a useful guide for both attack simulation and defensive system design.

2.2 Review of Existing Research

The use of machine learning and data from honeypots to improve cybersecurity in smart city internet of things (IoT) ecosystems was examined by Ahmed et al. [1]. The approach comprised using a variety of machine learning models, such as Naive Bayes, Decision Trees, K-Nearest Neighbours, Sequential Neural Networks, and long short-term memory (LSTM), on real-world attack data gathered from high interaction IoT honeypots over a period of 1.5 years. In order to validate the fundamental idea that AI-driven analysis of honeypot data is useful for IoT security, the study showed that the Decision Tree algorithm obtained accuracy above 99% throughout examination settings. Nevertheless, the approach relies on conventional machine learning models and ignores cutting-edge methods for sequence modelling, such Transformers, as well as multi-agent coordination for coordinated, active response.
Empirical data from a comparative study by Kaur et al. [3] provides supporting evidence for Transformer superiority over LSTM architectures. Transformer models considerably outperform LSTM-based techniques, achieving 89.19% accuracy and 89.51% F1-score compared to LSTM's 84.97% accuracy, according to the study's head-to-head comparison on text-based cyber threat classification tasks. The self-attention mechanism's higher performance is explained by its capacity to recognise intricate patterns and long-range interdependence.
For adaptive threat defence, Siddiqui et al. [10] suggested a multi-layered AI-driven cybersecurity architecture that combines game theory, fuzzy reasoning, entropy analytics, and multi-agent reinforcement learning. With a 97.24% detection accuracy, 2.18% false positive rate, and 96 ms response time, this work supports MARL as a cutting-edge method for coordinated cyber defence. The architecture outperforms four cutting-edge baselines, such as Federated Intrusion Detection and DeepDefenseNet. However, entropy analytics and fuzzy logic—statistical techniques that respond to abnormalities rather than comprehending attacker intent—are used by the system's detection and planning components.
Targeting the reconnaissance phase of Advanced Persistent Threats, Li et al. [2] discussed a multiple deception resources deployment technique utilising reinforcement learning for network threat mitigation. With a defensive success probability of 97.09%, the methodology used proximal policy optimisation (PPO) to produce clever deployment strategies for hybrid honeypot configurations. PPO's excellent combination of defence utility and convergence stability was demonstrated by the thorough study, which tested PPO against six RL algorithms (Q-Learning, DQN,

DDQN, and Actor-Critic). Nevertheless, the authors specifically point out two significant drawbacks: the method depends on a single RL agent dealing with computational restrictions, and the static deployment features find it difficult to adjust to later stages of an APT attack.

Using deep generative adversarial networks (DGANs) in conjunction with discrete hidden Markov models (DHMMs), Kumar et al. [11] presented an adaptive distributed honeypot detection network (ADHDN) for improved cybersecurity against denial of service (DoS) and distributed denial of service (DDoS) attacks. A central Honeypot Server Manager distributes traffic to several physical servers that house virtual machine detection agents as part of the distributed architecture. When compared to five modern baselines, experimental evaluation on the IoTID20 dataset showed remarkable true positive rates of 99.7% for protocol-level attacks with the lowest false positive rates. The authors clearly state that although the framework is very effective for DoS/DDoS detection, it needs to be extended to handle Advanced Persistent Threats and low-and-slow attacks.

An early intrusion detection system for industrial control networks employing honeypots was developed by Pashaei et al. [16], who also introduced a novel adversarial reinforcement learning training procedure using the state-action-reward-state-action (SARSA) algorithm. Two agents are used in the architecture: an environmental agent that deliberately chooses challenging attack instances to bolster the classifier and a classification agent that learns to identify intrusions. Using datasets from industrial control systems, this adversarial training achieved 99% accuracy and 0.98 F1-measure. The method is still a single-agent classification system that makes state-based judgements while validating RL for honeypot-based defence; it lacks deep sequential comprehension and multi-agent coordination capabilities.

A dynamic AI-powered honeypot platform that integrated Q-learning-based deception, machine learning detection (random forest, support vector machine (SVM)), real-time threat intelligence, and automated reaction mechanisms was introduced by Hussain et al. [9]. Compared to static honeypots, the proposed end-to-end framework increased attacker dwell time from 1.2 to 5.3 minutes which represented an improvement of nearly 342%, while achieving a detection accuracy of 97% with a 3.8% false positive rate. The adaptive strategy clearly outperformed the Cowrie and T-Pot baselines in every category. Nevertheless, without multi-agent coordinating capabilities, the system uses typical Q-learning for a single honeypot agent. Federated learning for distributed model updates is identified by the authors as future work.

Ragsdale and Boppana [5] examined the techniques for designing low risk honeypots using generative pre-trained transformer (GPT) models using curated inputs. Their approach makes the GPT-3.5-Turbo the back-end intelligence but with a novel Front-End Interface controlling context histories for an equilibrium between realism and token economies. Evaluation using Cowrie against five MITRE ATT&CK attack scenarios demonstrated significant increases in realism, and longer sustained deception of multi-step attacks, attaining 77% reduction in the number of tokens used as a result of intelligent context management. The LLM-based honeypot has a 0.0 score on Shodan's Honeyscore (perfect covertness). Whilst validating Transformer models for honeypot intelligence, the system focuses on response generation for one asset as opposed to using strategic coordination across multiple honeypots.

Wu et al. [7] proposed RTIDS, which is a robust Transformer-based intrusion de-

tection systems or provide fundamental advantages of transformer architecture for network security applications. The framework delivered outstanding F1-scores of CICIDS2017 99.17% and CIC-DDoS2019 98.48% significantly outperforming baselines support vector machine (SVM), recurrent neural network (RNN), feedforward neural network (FNN), and long short-term memory (LSTM). The architecture incorporates positional embeddings, stacked encoder–decoder layers, topological self-attention, and hierarchical reasoning. While the research strongly validates the suitability of Transformers for analyzing network security data, RTIDS remains a purely passive detection system without active deception capabilities, highlighting the potential for extending such models to adaptive honeypot defense.

Sun et al. [6] on honeypot identification for softwarized Industrial Cyber Physical Systems, considering the adversary point of view in order to comprehend which honeypot are detected. The methodology used included secure fuzzy testing to construct probe packets taking advantage of difference in error handling behavior 92% accurate in defending honeypots against real devices using convolutional neural network (CNN) classification. This research provides strong motivation for adaptive honeypots that can resist fingerprinting attempts, as static honeypots are inherently vulnerable to detection. The study clearly demonstrates the necessity of intelligent honeypots capable of realistic error handling and dynamic behavior in order to bypass sophisticated identification techniques.

Dumisani and Ould-Slimane [17] proposed a game theoretic approach to cyber deception based on redundancy of diversified honeypots. The methodology models attacker-defender conflict as a zero-sum game, working out mixed strategies that optimize them using Nash Equilibrium with linear programming. Simulation based on fictitious play algorithms showed significantly increasing the redundancy improves the defense rewards. While giving a solid theoretical base based on classical game theory, the approach calculates placement strategies to be applied before attacks occur, lacking the capabilities of a dynamic experiencing adaptation offered by reinforcement learning.

Shao et al. [12] have proposed a mimic honeypot construction method based on incomplete information of internet of things (IoT) security zero-sum stochastic games and Q-learning. The architecture is the implementation of Dynamic Heterogeneous Redundancy (DHR) with multiple business executors that are dynamically reconfigurable. The Q-learning algorithm derives Bayesian Nash equilibrium strategies, achieving an attack capture rate of 90%, while limiting virtual escape success to only 6%, compared to 50% for static honeypots. However, the system does not operate like many honeypots but as a single intelligent honeypot instead of a coordinated swarm.

Manocchio et al. [8] designed FlowTransformer, a comprehensive framework for the systematic analysis of Transformer network architectures for network intrusion detection. By exploring input encoding strategies in a broad grid search across Transformer configurations (including GPT-2, BERT, and shallow models) and classification heads, the study provides crucial architectural design guidance. Key findings 'Last Token' classification heads are significantly better than alternatives, and shallow Transformer models (2-4 layers) achieve comparable performance of larger models with the additional benefit of being faster and more efficient. These findings provide crucial architectural design guidance for Transformer-based network intrusion detection systems; the recommendation of shallow encoders with Last Token

heads provides an evidence-based basis for the design of the Transformer component.
Guan et al. [4] HoneyIoT, proposed HoneyIoT, an adaptive high-interaction honeypot for IoT devices based on reinforcement learning, which serves as the direct technological predecessor for multi-agent adaptive honeypots. The methodology consists of training RL agents (using PPO) on real-world attack traces gathered from vulnerable IoT devices, and combined with differential analysis for mutation of dynamically-generated content. Deployed on AWS for two months, HoneyIoT achieved remarkable results: 7.57 average session length versus 4.02 for conventional systems, collected 467 malware samples versus 14 for baselines, and maintained a Shodan Honeyscore of 0.0 (perfect covertness). Whilst demonstrating the power of single-agent adaptive honeypots, this work explicitly leaves the multiagent coordination problem open.
Abbas-Escribano and Hébar [15] Improving the model for honeypot attacks detection and analysis, changing the design philosophy from one based on vulnerability to process-centric approaches based on MITRE ATT&CK & Cyber Kill Chain frameworks. The proposed multi-layered architecture deploys honeypots at entry, intermediate, and internal levels, compelling attackers to progress through predefined stages. Deployment on the public internet for 17 days has attracted 1.5 millions events from 35000 unique IP addresses. While displaying sophisticated multi-stage design of deception, the entire attack path and vulnerabilities are previously determined by human experts the system does not learn or adapt.
Wushishi et al. [13] proposed D3O-IIoT, a deep reinforcement learning (DRL)-based dynamic deception orchestration framework for industrial IoT security. Using dueling deep Q-Network (dueling DQN), which is mimicking optimal policies for orchestrating a collection of deception tactics such as honeypot, IP shuffling, fake telemetry and node isolation. Explicitly balances attack mitigation, engagement, precision, and cost. Evaluation on the CIC-IIoT2025, WUSTL-IIoT2021, and TON-IoT datasets demonstrated an attack mitigation rate of 13.75% with only a 0.35% false alarm rate, along with a decision latency of 2.07 ms, thereby proving real-time feasibility. However, the architecture employs a single centralized DRL agent & state-based perception without sequence modeling.
Do Hoang et al. [18] introduced Hawkeyes, a hierarchical multi-agent reinforcement learning framework for intelligent honeypot allocation, achieving a 15–18% improvement in trapping efficiency compared to baseline RL methods. The two tier architecture uses a high level policy choosing strategies depending on stage of intrusion and the criticality of asset, coordinated with low level agents deciding precise honeypot placements. Network states are modelled as rich graphs of compromised nodes, topology, and vulnerabilities allowing stage aware decision making with 48.4% efficiency at 0.05 false negative rate (FNR) versus 30.6% for flat MARL. While showing a demonstration of the power of hierarchical multi-agent coordination for honeypot systems, the framework relates to graph-based feature extraction with no deep sequence modelling emphasizing the opportunity for Transformer based attacker intent understanding.
Chamotra and Barbhuiya [14] SAGE, Adaptive IoT honeypot Combinationing finite state machine (FSM)-powered protocol emulation with GraphRAG-powered LLM response generation. The hybrid architecture is based on Finite State Machines for standard protocol compliance and large language models (LLM) with graph retrieval augmented generation (GraphRAG) for processing unknown requests. More

than five month deployment recorded over double the interactions (12,320 vs 5,180) in comparison with Conpot and identified close to the unique attackers (1,750 vs 650). The system resulted in perfect protocol compliance showing ability to dynamically expand its FSM based on validated LLM responses. While representing state of art in single honeypot intelligence explicitly identify reinforcement learning for autonomous strategy optimization as future work.

2.3 Summary of Key Findings

The comprehensive literature review supports the proposed Transformer-MARL framework and identifies clear architectural requirements and research gaps:

Transformer Superiority for Sequential Analysis: Multiple studies demonstrate that Transformer architectures significantly have an edge out of LSTM with traditional models for cybersecurity sequence analysis, yielding F1-scores of more than 99% [3, 7]. A framework for designing transformers, the FlowTransformer framework [8], gives some advice on the design of shallow encoders with 'Last Token' classification heads giving the best performance efficiency trade-offs.

Reinforcement Learning Effectiveness for Adaptive Honeypots: Reinforcement learning based adaptive honeypots are far more successful, consistent, and accurate. HoneyIoT [4] collected 30 times more malware than traditional systems; Hussain et al. [9] increased the attacker dwell time by about 342%. These results validate ML (RL model) as the state-of-the-art way of adaptive deception.

Multi-Agent Coordination Gap: When single-agent adaptive honeypots demonstrate proven effectiveness [4, 9, 13], and multi-agent systems have the promise for cybersecurity [10, 18], No existing work integrates Transformer based sequence understanding using MARL for honeypot swarms coordination. This represents the primary research gap the proposed thesis will address.

Computational Feasibility: With recent systems it is proven that sophisticated AI driven defense can be used with real-time response constraints Wushishi et al. [13] achieved 2.07 ms decision latency and 96 ms response time from Siddiqui et al. [10] time. Those benchmarks provide proof of workable feasibility of Transformer-MARL integration.

Design Requirements from Literature: The review turns up certain technical requirements: (1) shallow Transformer encoders (use Last Token heads for efficiency) [8], (2) centralized training with decentralized execution (CTDE) architecture for multi-agent coordination [10], (3) multi-objective reward functions traded off engagement and cost of resources [13], and (4) MITRE Realistic evaluation of ATT&CK-based attack modeling [5, 15].

Chapter 3

Requirements, Impacts and Constraints

3.1 Final Specifications and Requirements

This section discusses technical or methodological Requirements, data or resource requirements (software), etc.

3.2 Societal Impact

Study and report on project impact on society, health, safety, legal, and cultural aspects.

3.3 Environmental Impact

Study and report on project sustainability and environmental impact.

3.4 Ethical Issues

Identify ethical issues and professional responsibilities related to the design solution.

3.5 Standards - if applicable

Write about any standard maintained.

3.6 Project Management Plan

Prepare a project management plan including schedule, budget, and resource management.

3.7 Risk Management

Identify project risks and mitigation strategies.

3.8 Economic Analysis

Conduct economic analysis and cost-benefit estimation.

Chapter 4

Proposed Methodology

4.1 Design Process and Methodology Overview

This chapter presents the complete architectural design, mathematical formulation, and implementation specification of the TEMARL framework for adaptive honeypot deception. The chapter is organized into four principal sections. Section 4.1 explains the overall design process, the main goals behind each architectural decision, and the important operational limitations that the framework must follow. Section 4.2 provides detailed model specifications for all five architectural components: the simulation environment and network topology, the Transformer-based attacker intent encoder, the Gated Multi-Modal Fusion Q-network, the QMIX monotonic mixing network, and the multi-objective reward function. Section 4.3 gives the end-to-end data-engineering pipeline through which 149 gigabytes of raw enterprise telemetry from the COMISET dataset are converted into GPU-ready training tensors, including all cleaning, transformation, integration, and reduction operations. Section 4.4 explains the three-phase training process, lists all hyperparameters and software requirements, and defines the combined training objective. Together, these sections give a complete and reproducible description of the proposed system, allowing others to independently verify all reported results.

4.1.1 Objectives and Constraints

There are two main goals that drive the design process. The first directly addresses the fingerprinting vulnerability of static defences found by Sun et al. by enabling a honeypot swarm to deduce attacker intent from the observed technique sequence and modify its false replies accordingly. [6]. Static honeypots' deterministic error-handling behaviour makes them useless for gathering intelligence since adversaries can use targeted probing packets to differentiate them from real hosts. [6]. In order to overcome the single-agent constraint specifically mentioned by Li et al., the second goal is to coordinate the swarm under a principled multi-agent framework such that the agents operate collectively rather than independently. [2], who point out that isolated agents encounter computing limitations and find it difficult to adjust to later phases of a multi-phase Advanced Persistent Threat (APT) campaign.
Four binding constraints shape every architectural decision. First, the system must operate under *partial observability*: network telemetry is delayed, incomplete, and potentially spoofed. This constraint motivated the POMDP formalisation intro-

duced in Section 4.1.2 and the fog-of-war evaluation protocol described in Chapter 5. Second, *real-time responsiveness* is required, comparable to the millisecond-scale responses reported in prior adaptive defences [10, 13], ensuring that the framework is viable for live deployment. Third, the design must be *fully reproducible* from a single locked random seed, enabling independent verification of every reported experimental result. Fourth, as will be covered in more detail in Section 4.2.3, the framework must be *computationally efficient*, operating solely on a single consumer GPU without the need for specialist cluster infrastructure, which limits the encoder depth and agent network size to compact configurations.

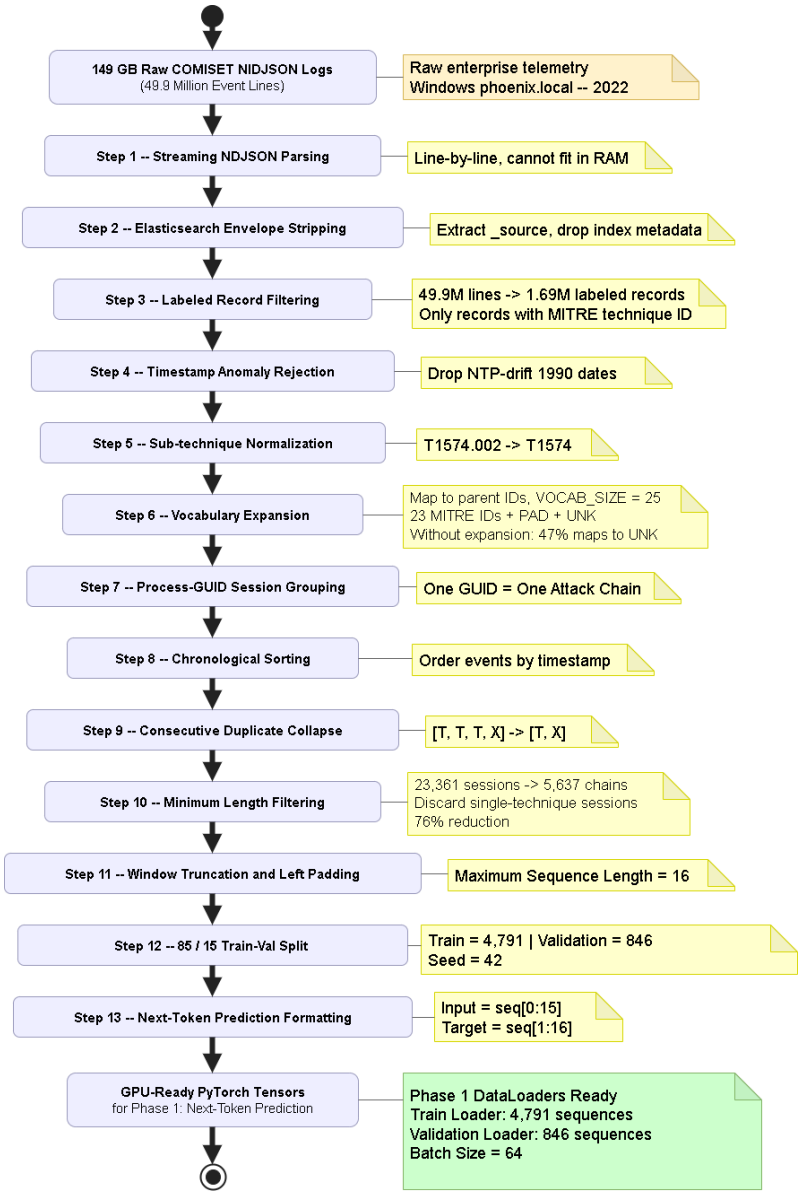


Figure 4.1: Runtime execution and feedback loop of the Transformer-Enhanced MARL framework.

4.1.2 Design Process

The execution-and-feedback loop in Figure 4.3 illustrates the runtime interaction between the four successive stages of the design. A cohesive end-to-end pipeline

from raw telemetry to coordinated deception operations is created by each stage addressing one of the 17 primary goals or limitations listed in Section 4.1.1. The output of each stage serves as the input to the subsequent stage.

1. **Problem formalisation.** The defensive job is formalised as a POMDP [10], where agents must behave in the face of uncertainty and perceive only a partial and perhaps distorted representation of the genuine environment state. The Transformer encoder, which deduces latent attacker intent from observable technique sequences in the absence of a directly accessible adversary state, and the fog-of-war evaluation protocol, which tests the learned policy when up to 90% of network telemetry is withheld to simulate realistic operational conditions, are both directly motivated by this framing. The POMDP framework recognises a basic fact of cyber defence: defenders operate with insufficient knowledge about attacker objectives, capabilities, and present position in the kill chain, even in the presence of dense sensor coverage.

2. **Architecture selection.** In Section 4.2.1, several possible architectures are assessed in relation to the goals and limitations. The chosen design incorporates an intent-conditioned QMIX mixing network, a Gated Multi-Modal Fusion Q-network, and a Transformer encoder into a Centralised Training, Decentralised Execution (CTDE) paradigm [10]. While each honeypot can operate autonomously at inference time utilising only its local observation and the common intent vector h, CTDE allows the central critic to access global state information and Transformer context during training. Scalability depends on this division of training and execution, which preserves the coordination advantages of joint training while avoiding the communication cost of fully centralised execution.

3. **Real-data grounding.** Using the thirteen-step data-engineering pipeline outlined in Section 4.3, the encoder is randomly initialised and trained from scratch using actual corporate telemetry from the COMISET dataset. The encoder would only be trained on artificial kill-chain sequences produced by the simulation in the absence of this grounding phase. These sequences adhere to a deterministic strategy ordering, which does not represent the intricate, non-sequential technique patterns seen in actual attacker campaigns. The sim-only encoder achieves zero Top-1 accuracy on held-out COMISET sessions, as shown in Chapter 5, demonstrating that fine-tuning on real data is not an optional upgrade but rather a requirement for generating an intent vector h that contains a true predictive signal.

4. **Training and evaluation.** The design is trained over a fixed computational budget with a locked random seed and evaluated against three baselines across four attacker strategies, a fog-of-war protocol spanning three telemetry obstruction levels, and a four-by-four cross-strategy generalisation matrix. This evaluation protocol is designed to expose both the strengths and the structural limitations of the framework, in accordance with the principle that honest assessment of failure modes is as scientifically valuable as the demonstration of successes.

4.2 Preliminary Design and Model Specification

4.2.1 Alternative Design Approaches and Selection

Before committing to the final architecture, three candidate designs were systematically evaluated against the objectives and constraints of Section 4.1.1. Table 4.1 summarises their characteristics and the rationale for rejection or selection.

	Alt. 1	**Alt. 2**	**Alt. 3 (Selected)**
Approach	Single-agent PPO.	LSTM encoder + independent DQN agents; no mixing.	Transformer encoder + Gated Fusion Q-net + QMIX.
Intent modelling	None.	Recurrent (LSTM).	Self-attentive Transformer [8, 19].
Coordination	None.	None (fully independent).	IGM-monotonic QMIX [20].
Partial-obs. robustness	Low.	Low.	High (gated fusion + state masking).
Verdict	Fails Objective 2 [10].	Lacks multi-head attention [8]; fails Objective 2.	Satisfies all objectives and constraints.

Table 4.1: Comparison of three candidate architectures against the design objectives and constraints.

Alternative 1 provides neither coordination nor intent modelling, making it incapable of collective action or attacker-state inference. Alternative 2 introduces recurrent intent modelling but relies on LSTM, which lacks the positional awareness and multi-head relational reasoning of the Transformer [8] and does not scale gracefully to long attack sequences. Furthermore, Alternative 2 trains agents independently with no mixing mechanism, preventing the IGM-condition guarantees required for principled cooperative Q-value decomposition [20]. In contrast, Alternative 3 simultaneously models intent through self-attention, coordinates agents through monotonic value factorisation, and remains robust to partial observability through gated fusion and training-time state masking. It is therefore selected for implementation in all subsequent sections.

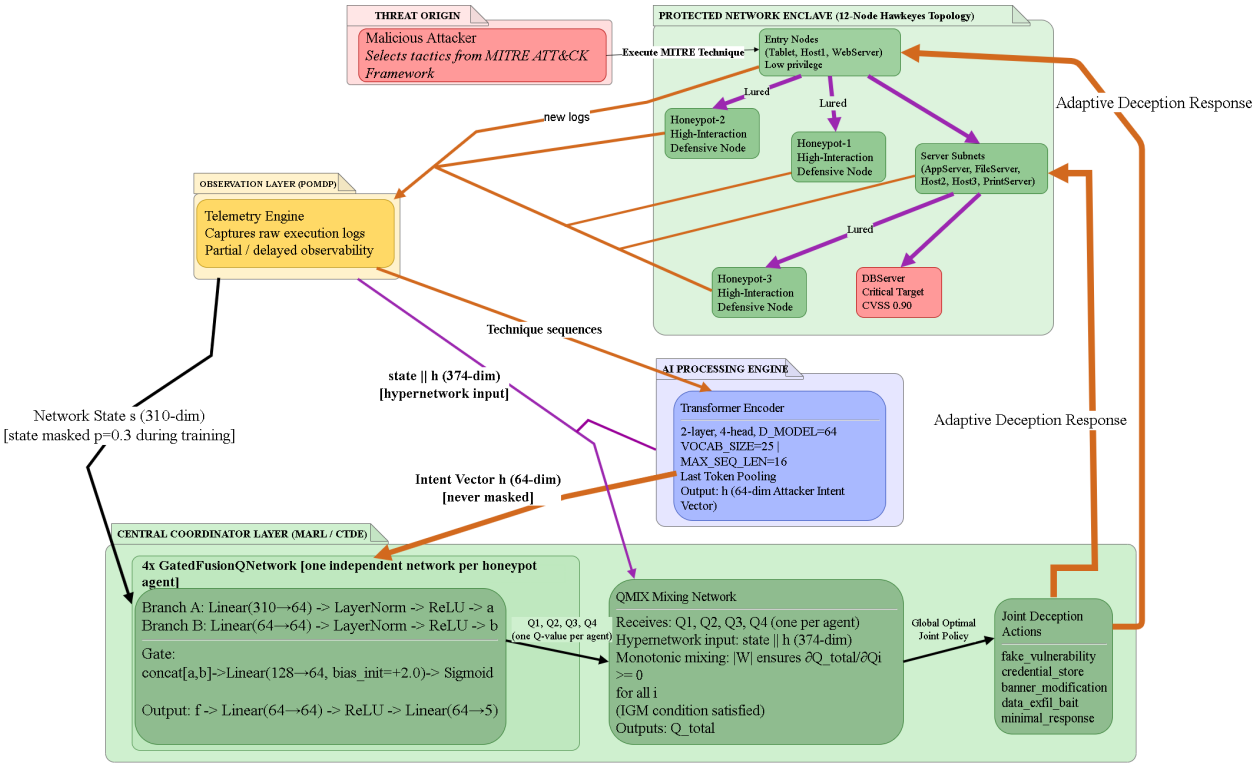


Figure 4.2: Macro-structural blueprint of the proposed Transformer-Enhanced MARL framework.

4.2.2 Simulation Environment

Figure 4.2 presents the macro-structural system blueprint of the proposed Transformer-Enhanced MARL Honeypot Deception Framework. Three processing layers are shown: the Protected Network Enclave (12-node Hawkeyes topology) at the top, the Observation and AI Processing Engine in the middle, and the Central Coordinator Layer (MARL/CTDE) at the bottom. The diagram is organised into three horizontal processing layers — the Protected Network Enclave at the top, the Observation and AI Processing Engine in the middle, and the Central Coordinator Layer at the bottom connected by directed data flows that collectively transform raw attacker behaviour into coordinated deceptive responses. Each block in the figure is described in detail below.

Threat Origin Block: The *Threat Origin* block (upper left) represents the adversarial agent in the simulation. The malicious attacker selects tactics from the MITRE ATT&CK framework and executes them against the protected network enclave via the *Execute MITRE Technique* dashed arrow. In the simulation, the attacker follows one of four behavioural strategies — defined in Table 4.4 — covering the full spectrum from highly structured kill-chains (HiE) to uniformly random technique selection (Ran). Each technique execution produces new log entries that feed the Observation Layer at the next simulation step.

Protected Network Enclave (12-Node Hawkeyes Topology): The twelve-node Hawkeyes business architecture of Do Hoang et al. [18], which offers a realistic multi-subnet network with heterogeneous node roles and a CVSS-based vulnerability score scheme, is replicated in the huge green area (upper right). Table 4.2 contains the whole node specs. There are three types of nodes in the enclave.

Entry Nodes (Tablet, Host1, WebServer) serve as the attacker's initial contact points with low privilege levels, forming the primary attack surface for Initial Access and reconnaissance techniques.
Server Subnets (AppServer, FileServer, Host2, Host3, PrintServer) represent genuine internal assets that the honeypot swarm must protect from lateral movement. The attacker seeks to traverse these nodes toward the critical target.
Honeypot Nodes (Honeypot-1, Honeypot-2, and Honeypot-3) are High-Interaction Defensive Nodes which are positioned at key subnet borders. The purple *Lured* arrows show that an attacker is drawn to a honeypot rather than real assets when its deception action achieves strong semantic alignment with the attacker's current tactic. Each of the three honeypots has a base CVSS score of 0.50, which is lowered by a deception multiplier of 0.6 to an effective attractiveness of 0.30, which is roughly 40% less appealing than the lowest-scoring genuine node. This ensures that engagement necessitates actual deceptive action rather than passive waiting.
DBServer (bottom right, highlighted in red) is the only topological path via PrintServer to the crucial target, which has a CVSS score of 0.90. The terminal reward penalty $R = -1.0$ is triggered when an attacker compromises DBServer, resulting in episode failure. The feedback loop from the coordinator layer back into the enclave is closed by the dashed brown *Adaptive Deception Response* arrows on the right, which show how emitted deception actions affect the attacker's routing choices at the subsequent stage.
Observation Layer (POMDP): The yellow *Observation Layer* block houses the Telemetry Engine, which captures raw execution logs from all nodes and assembles them into the 310-dimensional global state vector s. This vector is composed of three sub-matrices: the compromise matrix M_{com} (10 dimensions, binary compromise status per non-honeypot node), the connectivity matrix M_{con} (100 dimensions, 10×10 inter-node adjacency with compromised edges set to 0.8 to reflect reduced traversal reliability), and the vulnerability matrix M_{vul} (200 dimensions, twenty CVSS-scored vulnerability slots per node, perturbed with Gaussian noise $\sigma = 0.02$ to prevent overfitting to exact scores). The annotation *state masked p=0.3 during training* indicates that 30% of the 310 state dimensions are randomly zeroed at each training step to enforce robustness under partial observability; this masking is disabled at evaluation time. The label *Partial / delayed observability* acknowledges that in realistic deployments, telemetry may arrive out of order, be dropped, or be spoofed. At each step, the Observation Layer also passes the observed *Technique sequences* to the AI Processing Engine.
AI Processing Engine: The blue *AI Processing Engine* block (middle right) houses the Transformer Encoder, which converts the attacker's most recent sixteen-technique observation window into the 64-dimensional Intent Vector $h \in \mathbb{R}^{64}$. The encoder uses a 2-layer, 4-head architecture with D_MODEL=64, VOCAB_SIZE=25, MAX_SEQ_LEN=16, and Last Token Pooling; full parameters are specified in Table 4.7. The critical design decision captured by the annotation *Intent Vector h (64-dim) [never masked]* is that while s is subject to both training dropout and evaluation-time fog-of-war obstruction, h is always fully available to the agents. This asymmetry is the mechanistic explanation for the Transformer's greatest benefit manifesting under high telemetry obstruction: when the 310-dimensional state is heavily zeroed, h becomes the dominant source of actionable defensive signal. The concatenated vector state $\| h \in \mathbb{R}^{374}$, annotated as the *hypernetwork input*, is passed

downward to the QMIX Mixing Network.

Central Coordinator Layer (MARL / CTDE): The bottom green band implements the Centralized Training, Decentralized Execution paradigm through three sequential processing blocks.

4× GatedFusionQNetwork. Four independent Q-networks, one per honeypot agent, each receive both s and h and fuse them through the dual-branch gated architecture specified in Table 4.8. Branch A projects the 310-dimensional telemetry to 64 dimensions via Linear → LayerNorm → ReLU; Branch B projects the 64-dimensional intent vector identically. A learned sigmoid gate with bias initialised at +2.0 — yielding $\sigma(+2.0) \approx 0.880$ — weights telemetry at 88% and intent at 12% at training onset, stabilising early learning before the encoder has produced reliable signal. As training progresses and h becomes increasingly predictive, the gate shifts toward intent for structured attackers. Each network outputs five Q-values corresponding to the five deception actions in Table 4.5. During training, the 310-dimensional state is zeroed with probability $p = 0.3$ while h is never masked, forcing the development of an h-conditioned policy robust to telemetry loss.

QMIX Mixing Network. The four individual Q-values Q_1, Q_2, Q_3, Q_4 — one per agent — are combined into the team-level value Q_{total} using intent-conditioned hypernetworks specified in Table 4.9. Critically, the hypernetwork input is the concatenated vector state $\| h$ (374-dimensional) rather than state alone as in standard QMIX [20]. This makes the mixing weights W_1, b_1, W_2, b_2 functions of attacker intent, enabling intent-driven coordination: when h encodes credential-seeking behaviour, the mixer upweights the Q-value of the agent best aligned with `credential_store`. The monotonicity constraint $|W|$ ensures $\partial Q_{\text{total}}/\partial Q_i \geq 0$ for all agents i, satisfying the IGM condition required for CTDE.

Joint Deception Actions. The final output block lists the five actions emitted by the swarm: `fake_vulnerability`, `credential_store`, `banner_modification`, `data_exfil_bait`, and `minimal_response`, defined in full in Table 4.5. These are injected back into the protected network enclave as the Adaptive Deception Response, closing the feedback loop and completing one simulation step.

Information Flow Summary: Reading Figure 4.2 from top to bottom, the complete data path for one simulation step proceeds as follows: the attacker executes a MITRE ATT&CK technique; all nodes emit logs assembled by the Telemetry Engine into the 310-dimensional state s; simultaneously, the Transformer Encoder converts the observed technique history into the 64-dimensional intent vector h; each GatedFusionQNetwork fuses s and h to produce per-agent Q-values; the QMIX Mixing Network combines these under the IGM constraint into Q_{total}; and the globally optimal joint deception action is emitted and fed back into the enclave as the Adaptive Deception Response.

ID	Name	Role	Subnet	CVSS	Adjacency
0	Tablet	Entry Point	1	0.72	$\{1, 9\}$
1	Host1	Entry Point	1	0.68	$\{0, 2, 9\}$
2	WebServer	Entry Point	2	0.85	$\{1, 3, 10\}$
3	AppServer	Internal	2	0.74	$\{2, 4, 5\}$
4	FileServer	Internal	2	0.61	$\{3, 10\}$
5	Host2	Internal	3	0.69	$\{3, 6, 11\}$
6	Host3	Internal	3	0.71	$\{5, 7, 11\}$
7	PrintServer	Internal	3	0.58	$\{6, 8\}$
8	DBServer	Critical Target	4	0.90	$\{7\}$
9	Honeypot1	Honeypot	1	0.50	$\{0, 1\}$
10	Honeypot2	Honeypot	2	0.50	$\{2, 4\}$
11	Honeypot3	Honeypot	3	0.50	$\{5, 6\}$

Table 4.2: Network node inventory of the twelve-node Hawkeyes topology [18].

The network node inventory is presented in Table 4.2. The twelve nodes are distributed across four subnets, with nodes 0, 1, and 2 serving as entry points in subnets 1 and 2, and node 8 (DBServer) serving as the critical target in subnet 4. The three honeypot nodes (9, 10, 11) are strategically placed at subnet boundaries to intercept lateral movement: Honeypot1 covers the entry subnet, Honeypot2 sits at the boundary between subnets 2 and 3, and Honeypot3 guards the approach to the internal subnet preceding the critical target. CVSS values represent base vulnerability scores. All three honeypots are assigned a CVSS score of 0.50, which is then multiplied by a deception factor of 0.6, yielding an effective attractiveness of 0.30. This reduction makes honeypots approximately forty percent less attractive than the lowest-scoring genuine node (PrintServer at 0.58), ensuring that a purely greedy attacker would not preferentially target them, and that engagement requires genuine deceptive action on the honeypots' part. Consequently, the alignment matrix of Table 4.6 becomes the central training signal: the only way for honeypots to increase engagement is to present actions whose semantic fit with the current attacker technique is high.

The most striking finding in Table 4.2 is the stark difference between the entry points and the critical target: The three honeypots all score 0.50 at baseline, but DBServer has the highest CVSS score (0.90), making it the most appealing node to a rapacious attacker. WebServer (0.85) forms the second-most attractive legitimate target and serves as the primary entry vector for exploitation-focused attackers, explaining the high alignment between Initial Access techniques and `banner_modification` and `fake_vulnerability` actions in Table 4.6. Moreover, the single adjacency edge between PrintServer (node 7) and DBServer (node 8) creates a topological bottleneck that concentrates the attacker's lateral-movement path, amplifying the strategic importance of Honeypot3's placement in subnet 3.

ID	Code	Name	Tactic	Source
0	T1190	Exploit Public-Facing Application	Initial Access	Sim
1	T1566	Phishing	Initial Access	Sim
2	T1059	Command and Scripting Interpreter	Execution	Sim
3	T1204	User Execution	Execution	Sim
4	T1053	Scheduled Task/Job	Persistence	Sim
5	T1543	Create or Modify System Process	Persistence	Sim
6	T1547	Boot or Logon Autostart Execution	Persistence	Sim
7	T1068	Exploitation for Privilege Escalation	Privilege Escalation	Sim
8	T1055	Process Injection	Privilege Escalation	Sim
9	T1046	Network Service Scanning	Discovery	Sim
10	T1016	System Network Config Discovery	Discovery	Sim
11	T1083	File and Directory Discovery	Discovery	Sim
12	T1021	Remote Services	Lateral Movement	Sim
13	T1550	Use Alternate Auth Material	Lateral Movement	Sim
14	T1072	Software Deployment Tools	Lateral Movement	Sim
15	T1005	Data from Local System	Collection	Sim
16	T1560	Archive Collected Data	Collection	Sim
17	T1041	Exfiltration Over C2 Channel	Exfiltration	Sim
18	T1486	Data Encrypted for Impact	Impact	Sim
19	T1489	Service Stop	Impact	Sim
20	T1036	Masquerading	Defense Evasion	COMISET
21	T1574	Hijack Execution Flow	Persistence	COMISET
22	T1553	Subvert Trust Controls	Defense Evasion	COMISET
23	PAD	—	Special	Token
24	UNK	—	Special	Token

Table 4.3: MITRE ATT&CK technique vocabulary (VOCAB_SIZE = 25) [15]. Tokens 0–19 are used by the simulation attacker; tokens 20–22 were added after COMISET analysis; tokens 23–24 are special tokens.

The scripted attacker draws techniques from a 25-token MITRE ATT&CK vocabulary described in Table 4.3. Tokens 0–19 form the core kill-chain vocabulary used by the simulation attacker, covering six tactic phases from Initial Access through Impact. Tokens 20–22 were added following COMISET analysis, which revealed that three techniques — Masquerading (T1036), Hijack Execution Flow (T1574), and Subvert Trust Controls (T1553) — together account for approximately fifty-seven percent of all labelled records in the real enterprise dataset and would otherwise map to the UNK token, producing information loss. The vocabulary expansion from 22 to 25 tokens therefore serves a dual purpose: it preserves semantic fidelity during fine-tuning and enables the encoder to generalise from the dominant patterns in real attacker behaviour to the more structured kill-chain sequences of the simulation.

Code	Name	Description	Predictability
HiE	High-Exploitation	Follows a tightly ordered kill chain with ≈90% exploitation actions.	High
HiC	High-Credential	Prioritises credential access and lateral movement (T1550, T1072).	Moderate
Ran	Random	Samples uniformly at random from IDs 0–19.	None
Mix	Mixed	Blends exploitation, credential, discovery, and exfiltration behaviours.	Moderate

Table 4.4: Four scripted attacker strategies ordered by decreasing sequence predictability. Predictability level directly governs the benefit the Transformer encoder can provide.

The four scripted attacker strategies presented in Table 4.4 span the full spectrum of behavioural predictability. The High-Exploitation (HiE) strategy follows a tightly ordered kill chain in which approximately ninety percent of technique selections are

exploitation-oriented, producing highly structured sequences that the Transformer encoder can model with strong predictive accuracy. At the opposite extreme, the Random (Ran) strategy samples uniformly from IDs 0–19, producing sequences that contain no exploitable temporal structure and represent an important structural limitation boundary for the proposed architecture, as discussed in Chapter 5. The High-Credential (HiC) and Mixed (Mix) methods are in the middle, with Mix combining several strategy classes into somewhat predictable sequences and HiC focusing on lateral-movement and credential-access techniques (T1550, T1072). A thorough and truthful description of the system's capabilities is made possible by this spectrum, which guarantees that the evaluation encompasses both the domain of maximal Transformer benefit and the region where the architecture's assumptions are broken.

ID	Name	Purpose and Semantic Target
0	`minimal_response`	Default low-engagement response; no active deception. Serves as the null-action baseline.
1	`banner_modification`	Alters service banners to match the attacker's active reconnaissance interest, drawing the attacker into further interaction.
2	`fake_vulnerability`	Presents an artificial exploitable vulnerability; also the static baseline policy default.
3	`credential_store`	Deploys a fabricated credential repository designed to bait credential-theft and lateral-movement stage attackers.
4	`data_exfil_bait`	Plants attractive fake data collections to intercept and prolong attacker engagement during Collection and Exfiltration phases.

Table 4.5: Honeypot deception action space (five discrete actions per agent) [5, 9].

The five honeypot deception actions are defined in Table 4.5. Action 0 (`minimal_response`) is the default null-deception baseline, producing no active engagement signal. Action 2 (`fake_vulnerability`) is also the static honeypot baseline used throughout Chapter 5, because a fixed-action policy that always presents a visible vulnerability represents the most common conventional honeypot deployment. The remaining three actions are semantically targeted: `banner_modification` is most effective against reconnaissance-phase attackers (ID 9–11), `credential_store` is optimally aligned with lateral-movement techniques (ID 12–14), and `data_exfil_bait` is designed for Collection and Exfiltration techniques (ID 15–17). The design follows the dynamic deception action families established in prior adaptive honeypot literature [5, 9].

The semantic relationship between attacker techniques and honeypot actions is quantified in the 23×5 alignment matrix presented in Table 4.6. Each of the 115 entries encodes a fitness score in the range $[0.0, 1.0]$ representing how well a given deception action satisfies the informational and operational expectations of an attacker executing a particular technique. These scores were specified by design to reflect established adversary behaviour models [15] and serve dual roles: as the primary learning signal in the reward function (via the reactive alignment component r_2 described in Section 4.2.6) and as the primary evaluation metric in Chapter 5.

An examination of Table 4.6 reveals several structurally important patterns. First, the minimum response action (column 1) achieves universally low scores through-

out the entire matrix, ranging from 0.1 to 0.2, confirming that a passive honeypot provides minimal semantic engagement regardless of the attacker's technique. This finding validates the design choice to penalise passive behaviour through the reward function. Second, the `banner_modification` action (column 2) achieves its highest values for Discovery-phase techniques: T1016 (System Network Config Discovery) reaches the matrix maximum for this action at 0.9, T1046 (Network Service Scanning) scores 0.8, and T1083 (File and Directory Discovery) scores 0.7. Consequently, a honeypot facing a reconnaissance-focused attacker should select `banner_modification` to maximise engagement; any other action at this phase reduces alignment by up to 0.8 points. Third, the `fake_vulnerability` action (column 3) dominates during Initial Access and Privilege Escalation: T1190 (Exploit Public-Facing Application) achieves the overall matrix maximum of 1.0 for this action, and T1068 (Exploitation for Privilege Escalation) scores 0.8. Nevertheless, this action performs badly during Lateral Movement and Exfiltration, with values as low as 0.2, indicating that static honeypots that solely rely on more than half of the kill chain is out of alignment with `fake_vulnerability` (the static baseline in Chapter 5). Fourth, the `credential_store` action (column 4) peaks for Lateral Movement techniques: T1550 (Use Alternate Auth Material) scores 0.6 and T1072 (Software Deployment Tools) reaches the column maximum of 0.9, indicating a strong semantic fit between lateral movement and credential manipulation. Additionally, the `data_exfil_bait` action (column 5) forms a tight cluster of high-alignment scores at the Collection-Exfiltration boundary, reaching its maximum of 1.0 for T1041 (Exfiltration Over C2 Channel), scoring 0.9 for T1005 (Data from Local System), and 0.8 for T1560 (Archive Collected Data). Overall, the matrix encodes a clear story: in order for deception to be effective, the attacker's current kill-chain phase must be semantically aligned. The Transformer encoder's job is to precisely infer this phase from the observed technique sequence so that agents can choose the action that is most aligned.

ID	Technique	min	banner	fake_v	cred	exfil
0	T1190 Exploit Public-Facing	0.1	0.4	1.0	0.2	0.3
1	T1566 Phishing	0.1	0.6	0.7	0.1	0.1
2	T1059 Command Interpreter	0.2	0.3	0.5	0.3	0.4
3	T1204 User Execution	0.1	0.5	0.4	0.2	0.3
4	T1053 Scheduled Task	0.2	0.3	0.4	0.4	0.3
5	T1543 Create System Process	0.1	0.4	0.5	0.3	0.3
6	T1547 Boot Autostart	0.1	0.5	0.4	0.3	0.2
7	T1068 Privilege Escalation	0.1	0.2	0.8	0.4	0.3
8	T1055 Process Injection	0.1	0.2	0.6	0.5	0.3
9	T1046 Network Scanning	0.2	0.8	0.5	0.1	0.2
10	T1016 Network Config Disc.	0.2	0.9	0.4	0.1	0.1
11	T1083 File/Dir Discovery	0.2	0.7	0.6	0.1	0.2
12	T1021 Remote Services	0.1	0.5	0.6	0.7	0.4
13	T1550 Alt Auth Material	0.1	0.4	0.5	0.6	0.4
14	T1072 Software Deploy Tools	0.1	0.3	0.4	0.9	0.3
15	T1005 Data from Local System	0.1	0.2	0.3	0.5	0.9
16	T1560 Archive Collected Data	0.1	0.2	0.3	0.4	0.8
17	T1041 Exfil Over C2	0.1	0.1	0.2	0.3	1.0
18	T1486 Data Encrypted	0.1	0.2	0.6	0.3	0.5
19	T1489 Service Stop	0.2	0.3	0.5	0.2	0.4
20	T1036 Masquerading	0.2	0.5	0.7	0.4	0.3
21	T1574 Hijack Exec Flow	0.1	0.4	0.6	0.5	0.3
22	T1553 Subvert Trust	0.2	0.6	0.7	0.3	0.2

Table 4.6: Full alignment matrix (23×5).

Each element represents the semantic fitness score in $[0.0, 1.0]$ between a honeypot

deception action (columns) and an attacker tactic (rows). Stronger semantic alignment is indicated by higher values. Every remaining action peaks at a different kill-chain phase, and the minimum response (min) is always low.

4.2.3 Transformer Encoder

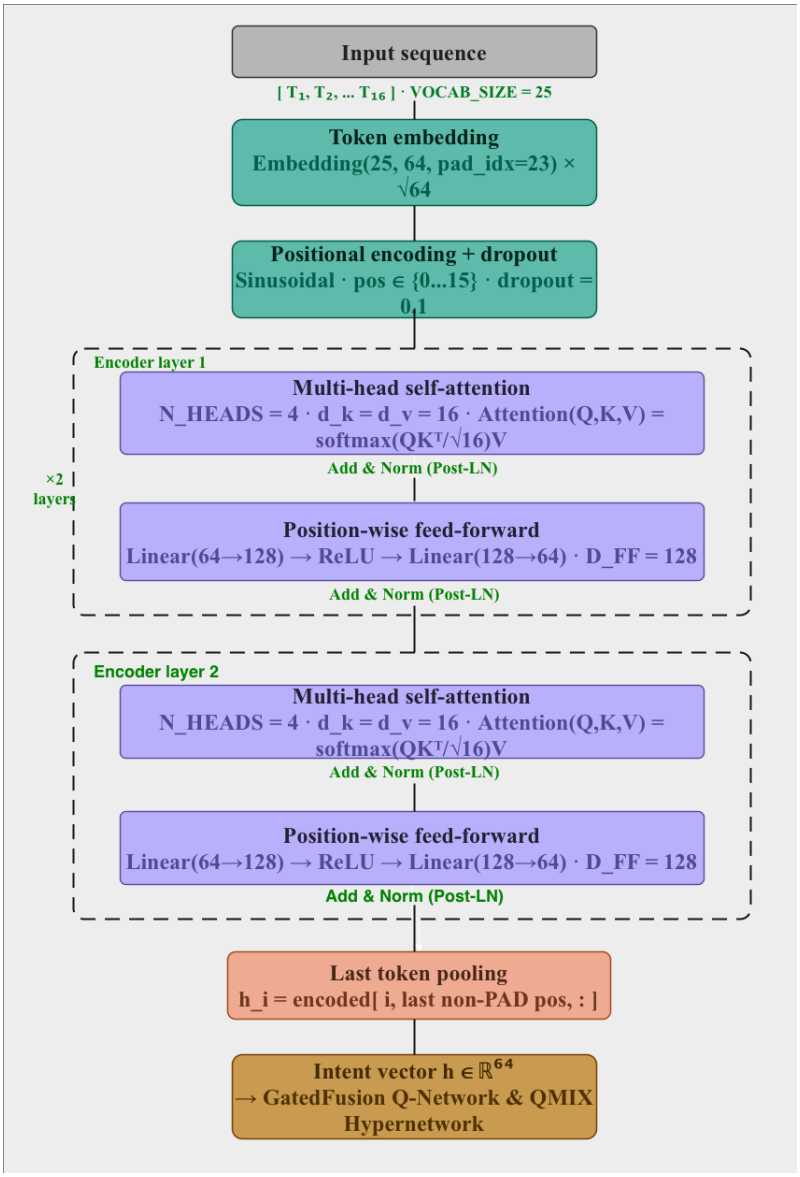


Figure 4.3: Transformer.

The Transformer encoder converts the attacker's most recent sixteen-technique window into the intent vector $h \in \mathbb{R}^{64}$, following the shallow last-token-pooling design of FlowTransformer [8] and the foundational Transformer architecture of Vaswani et al. [19]. The encoder architecture parameters are summarised in Table 4.7. The architectural choices reflect several deliberate engineering trade-offs that are justified in detail below.

Parameter	Value	Notes and Justification
VOCAB_SIZE	25	20 simulation techniques + 3 COMISET extensions + PAD (23) + UNK (24).
D_MODEL	64	Embedding and hidden dimension; compact but sufficient for 25-token vocabulary.
N_HEADS	4	$d_k = d_v = 16$ per head; 4 heads preferred over 8 to avoid head-collapse on small vocabulary.
N_LAYERS	2	Shallow stack per FlowTransformer [8]; two layers match larger models with lower cost.
D_FF	128	Feed-forward inner dimension = 2× D_MODEL; standard ratio.
DROPOUT	0.1	Applied after positional encoding and within the FFN sub-layer.
MAX_SEQ_LEN	16	Sliding window over most recent 16 techniques.
Pooling	Last Token	Final non-PAD position; shown superior by FlowTransformer [8].
Output	$h \in \mathbb{R}^{64}$	Supplied to both the Q-network and the QMIX hypernetwork.

Table 4.7: Transformer encoder parameters (AttackerIntentEncoder).

The choice of four attention heads ($N_HEADS = 4$) rather than eight deserves explicit justification. Eight heads would allocate just eight dimensions per head ($d_k = d_v = 8$) with a model dimension of 64, getting closer to the regime where individual heads operate on too few dimensions to learn unique relational patterns. On the other hand, each of the four heads allots 16 dimensions, which is enough to acquire different positional and tactic-transition associations while maintaining a consistent total number of parameters per attention layer. This logic adheres to the advice of Vaswani et al. [19], who point out that in order to prevent representation collapse, head count should be balanced against d_k.

The shallow two-layer architecture ($N_LAYERS = 2$) is motivated by the empirical finding of FlowTransformer [8] that shallow Transformer models with two to four layers achieve performance comparable to larger models on network-security sequence classification tasks, while offering substantially lower training and inference cost. For a vocabulary of twenty-five tokens and sequences of maximum length sixteen, a two-layer encoder is sufficient to capture inter-tactic dependencies across the full kill-chain window. Deeper encoders would risk overfitting on the limited COMISET dataset (5,637 sessions) and would violate the computational efficiency constraint.

The three core mathematical operations of the encoder are defined in Equations 4.1–4.3.

$$\mathrm{PE}(pos, 2i) = \sin\left(\frac{pos}{10000^{2i/d_{\text{model}}}}\right), \quad \mathrm{PE}(pos, 2i{+}1) = \cos\left(\frac{pos}{10000^{2i/d_{\text{model}}}}\right) \tag{4.1}$$

In Equation 4.1, $pos \in \{0, \ldots, 15\}$ is the position of the technique token within the sixteen-step sliding window, $i \in \{0, \ldots, 31\}$ is the dimension index within the 64-dimensional embedding, and $d_{\text{model}} = 64$ is the model hidden dimension. Sinusoidal positional encoding is chosen over learned positional embeddings because it provides a deterministic injection of positional order without additional trainable parameters, and because it generalises gracefully to sequence positions not seen during training [19].

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^\top}{\sqrt{d_k}}\right)V, \quad \text{MultiHead} = \text{Concat}(\text{head}_1, \ldots, \text{head}_h)\, W^O \tag{4.2}$$

In Equation 4.2, Q, K, V are the query, key, and value matrices respectively, computed from the embedded token sequence by learned linear projections. The scalar $d_k = 16$ is the per-head key dimension, and the scaling by $\sqrt{d_k}$ prevents the dot-product from saturating the softmax into near-zero gradients in high-dimensional spaces [19]. The multi-head formulation with $h = 4$ heads enables the model to jointly attend to information from different representational subspaces at different positions, capturing both local tactic transitions and longer-range kill-chain dependencies simultaneously.

$$h_i = \text{encoded}\Big[i,\ \arg\max_j \left(\text{token_ids}[i, j] \neq \text{PAD}\right),\ :\Big] \tag{4.3}$$

Equation 4.3 defines Last Token Pooling: for each sequence in the batch (index i), the pooled representation is taken from the final non-padding position, identified as the arg max over all positions j where the token identifier differs from the PAD token ID (23). This pooling strategy was identified as superior to mean pooling and CLS-token pooling by FlowTransformer [8], because it naturally captures the most recent attacker state — the position in the kill chain at the time of inference — without diluting the representation with early-phase information that may no longer be relevant. An all-PAD guard applied at inference time replaces the final position with T1190 whenever the entire window contains only padding, ensuring that Equation 4.3 always resolves to a valid, non-degenerate position at the start of an episode.

The forward pass proceeds through seven tensor operations: (1) token embedding via Embedding(25, 64, pad_idx=23) scaled by $\sqrt{64}$; (2) sinusoidal positional encoding added and dropout applied; (3) application of the first encoder layer comprising multi-head self-attention followed by Add & Norm and then position-wise feed-forward ($64 \to 128 \to 64$) followed by Add & Norm with Post-LN normalisation (`norm_first=False` in PyTorch); (4) application of the second encoder layer identically; (5) last-token pooling via Equation 4.3 to produce the output $h \in \mathbb{R}^{64}$; (6) passing h to the pre-training prediction head (Linear($64 \to 25$)) during supervised pre-training; and (7) passing h to the Q-network and QMIX hypernetwork during reinforcement-learning training.

4.2.4 Gated Multi-Modal Fusion Q-Network

Component	Specification
Branch A (telemetry)	Linear($310 \to 64$) $\to$ LayerNorm(64) $\to$ ReLU $\to a \in \mathbb{R}^{64}$.
Branch B (intent)	Linear($64 \to 64$) $\to$ LayerNorm(64) $\to$ ReLU $\to b \in \mathbb{R}^{64}$.
Gate	$\text{concat}[a, b] \in \mathbb{R}^{128} \to \text{Linear}(128 \to 64,\ \text{bias}_{\text{init}} = +2.0) \to$ Sigmoid $\to g \in (0, 1)^{64}$.
Fusion	$f = g \odot a + (1 - g) \odot b \in \mathbb{R}^{64}$.
Output head	$f \to$ Linear($64 \to 64$) $\to$ ReLU $\to$ Linear($64 \to 5$) $\to Q \in \mathbb{R}^5$.
State masking (train)	310-dim state zeroed with $p = 0.3$; h never masked. Forces h-conditioned policy development.
NoTrans ablation	$h = \mathbf{0}_{64}$ at all steps; h carries no signal. Functional, not structural, ablation.

Table 4.8: Gated Multi-Modal Fusion Q-network architecture (per agent, ≈37k parameters).

Each of the three honeypot agents maintains an independent Q-network that fuses the 310-dimensional network telemetry and the 64-dimensional intent vector h into per-action Q-values over the five-element deception action space. The fusion architecture, summarised in Table 4.8, is designed to address a structural imbalance inherent in naive concatenation: if the 374-dimensional combined input were fed directly to a single linear layer, the telemetry component would occupy $310/374 \approx 83\%$ of the input width, structurally suppressing gradients flowing from h and effectively preventing the encoder's signal from influencing the policy. The dual-branch architecture corrects this imbalance by projecting both modalities to the same 64-dimensional space before fusion, equalising gradient magnitudes.

$$g = \sigma(W_g \cdot \text{concat}[a, b] + b_g), \quad f = g \odot a + (1 - g) \odot b, \quad Q = W_2 \cdot \text{ReLU}(W_1 \cdot f) \tag{4.4}$$

In Equation 4.4, $a \in \mathbb{R}^{64}$ is the ReLU-activated LayerNorm output of Branch A (telemetry), $b \in \mathbb{R}^{64}$ is the corresponding output of Branch B (intent), $W_g \in \mathbb{R}^{64 \times 128}$ and $b_g \in \mathbb{R}^{64}$ are the gate projection weight and bias, σ is the element-wise sigmoid, and $g \in (0, 1)^{64}$ is the learned gate vector. The element-wise products $g \odot a$ and $(1 - g) \odot b$ represent the telemetry and intent contributions respectively, and their sum f is the fused representation passed to the output head. $W_1 \in \mathbb{R}^{64 \times 64}$ and $W_2 \in \mathbb{R}^{5 \times 64}$ are the output head weights. The gate bias b_g is initialised to $+2.0$, so that $\sigma(+2.0) \approx 0.880$ at the start of training. This means agents initially weight telemetry at 88% and intent at 12%, which stabilises early training by preventing the uninformative randomly-initialised encoder from disrupting Q-value estimates before the encoder has been sufficiently updated. As training progresses and h becomes increasingly predictive, the gate learns to shift toward intent for structured attackers.

LayerNorm is applied in both branches to normalise the scale difference between the 310-dimensional telemetry projection (whose input components span the range $[0, 1]$ for CVSS scores and binary compromise flags) and the 64-dimensional intent vector (whose components emerge from a trained encoder with a different distributional range). Without normalisation, one branch could dominate purely due to scale rather than informational content, corrupting the gate signal. The state masking

probability of $p = 0.3$ during training serves a complementary role: by randomly zeroing the telemetry state with thirty-percent probability, the training procedure forces agents to develop a robust h-conditioned policy that does not rely entirely on telemetry availability. This directly addresses the partial-observability constraint and is the mechanistic explanation for the Transformer's advantage under the fog-of-war evaluation.

4.2.5 QMIX Mixing Network

The QMIX mixing network combines the three individual agents' Q-values into a single team-level value Q_{total} while enforcing the Individual-Global-Max (IGM) monotonicity condition that guarantees the decentralised greedy action selection of each agent is globally optimal [20]. The architecture specification is presented in Table 4.9. The key architectural innovation relative to standard QMIX [20] is the inclusion of the intent vector h in the hypernetwork input: whereas the original formulation conditions the mixing weights exclusively on the global state, the proposed design concatenates state$\|h \in \mathbb{R}^{374}$ as the hypernetwork input. This makes the mixing weights W_1, b_1, W_2, b_2 functions of attacker intent, enabling intent-driven coordination: when h encodes a credential-seeking behaviour pattern, the mixer can upweight the Q-value of the agent whose policy is most aligned with `credential_store`, producing intent-responsive swarm coordination that is not achievable with state-only hypernetworks.

Component	**Specification**
Hyper input	state $\| h \in \mathbb{R}^{374}$ (standard QMIX uses state only [20]).
Hyper-W1	Linear($374 \to 128$) $\to$ ReLU $\to$ Linear($128 \to 256$) $\to$ reshape(4×64).
Hyper-b1	Linear($374 \to 64$).
Hyper-W2	Linear($374 \to 128$) $\to$ ReLU $\to$ Linear($128 \to 64$) $\to$ reshape(64×1).
Hyper-b2	Linear($374 \to 64$) $\to$ ReLU $\to$ Linear($64 \to 1$).
Mixing	$Q_{\text{total}} = \lvert W_2 \rvert \cdot \text{ELU}(\lvert W_1 \rvert \cdot Q_{\text{agents}} + b_1) + b_2$.
Monotonicity	abs(W) on W_1, W_2 guarantees $\partial Q_{\text{total}} / \partial Q_i \geq 0 \; \forall i$ [20].

Table 4.9: QMIX mixing network (QMIXMixer). The critical innovation is the inclusion of h in the hypernetwork input, enabling intent-aware mixing weight adaptation.

$$[W_1, b_1, W_2, b_2] = \text{HyperNet}(\text{state} \,\|\, h), \quad Q_{\text{total}} = |W_2| \cdot \text{ELU}(|W_1| \cdot Q_{\text{agents}} + b_1) + b_2 \tag{4.5}$$

In Equation 4.5, $W_1 \in \mathbb{R}^{4\times64}$ and $W_2 \in \mathbb{R}^{64\times1}$ are the intent-conditioned mixing weight matrices, $b_1 \in \mathbb{R}^{64}$ and $b_2 \in \mathbb{R}$ are the corresponding bias vectors, and $Q_{\text{agents}} \in \mathbb{R}^4$ is the vector of individual agent Q-values. The absolute-value operation applied to W_1 and W_2 enforces the monotonicity condition $\partial Q_{\text{total}} / \partial Q_i \geq 0$ for all agents i, which is the IGM condition required to ensure that the centralised Q_{total} and the decentralised greedy actions are mutually consistent [20]. Without this restriction, the CTDE guarantee could be compromised by agents learning locally optimum but globally suboptimal actions. Additionally, the ELU activation in the first mixing layer is better than ReLU because it allows negative values. This means that when individual agents deviate from the collectively preferred strat-

egy, the mixer can lower Q_{total} and produce a stronger coordination signal than a non-negative activation would.

4.2.6 Reward Function

The design philosophy of D3O-IIoT [13], which explicitly balances engagement, precision, and operational cost in a weighted linear combination, is adhered to by the multi-objective reward function. Equation 4.6 provides the complete definition of reward.

$$\begin{aligned} R &= 0.3\,r_1 + 0.40\,r_2 + 0.2\,r_3 + 0.1\,r_4 + 0.2\,r_5 \quad \text{(Proposed model)} \\ R &= 0.3\,r_1 + 0.40\,r_2 + 0.2\,r_3 + 0.1\,r_4 \quad \text{(NoTrans ablation)} \end{aligned} \tag{4.6}$$

The following is a definition of the five reward components. When the attacker is positioned at any honeypot node, the engagement reward $r_1 \in \{0, 1\}$ is a binary signal that is equal to 1.0; otherwise, it is equal to 0.0. This binary signal serves as a sparse but clear indicator of successful interception. The most significant learning signal is the reactive alignment reward $r_2 \in [0.1, 1.0]$, which is calculated as the maximum of `ALIGNMENT_MATRIX`[tech][a] over all agent actions a at the current step. It drives agents toward contextually appropriate deception by directly optimising the semantic fit between the attacker's technique and the honeypot's response. To reflect this centrality, r_2 is given the greatest weight of 0.40. The fraction of distinct actions among the four agents at each phase is the coordination reward $r_3 \in [0.25, 1.0]$, which encourages diversification, keeps agents from collapsing to a single action, and guarantees that the swarm as a whole covers the entire action space. When combined with r_3, the efficiency penalty $r_4 \in [-0.3, 0.0]$ offers both a positive incentive for diversity and a negative penalty for redundancy. It is calculated as $-0.1 \times (n_{\text{agents}} - \text{unique_actions})$, adding a small negative contribution when agents produce redundant actions. Lastly, when the encoder's anticipated next method matches the actual technique seen at the present step, the proactive bonus $r_5 \in \{0, 1\}$ is a binary signal equal to 1.0. In order to avoid an effective doubling of the alignment weight when predictions are accurate, which would happen if the bonus were folded into r_2, this component is purposefully separated from r_2. Since the $h = \mathbf{0}$ signal yields no significant prediction, the NoTrans ablation completely eliminates r_5. When the attacker successfully accesses the DBServer node, a terminal penalty of $R = -1.0$ is applied, giving a strong negative signal for policy failure.

4.3 Data Collection and Preprocessing

Real enterprise telemetry from the COMISET dataset [21] is used to fine-tune the Transformer encoder described in Section 4.2.3 after it has been pre-trained on synthetic attack sequences produced by the simulation environment. Because the simulated attacker employs a deterministic tactic ordering that differs from the intricate, non-sequential approach patterns seen in real infiltration campaigns, the fine-tuning step is crucial. COMISET is made up of about 49.9 million event lines in 149 gigabytes of raw newline-delimited JSON (NDJSON) event logs that were gathered from a Windows enterprise environment. A systematic thirteen-step pipeline is needed

to transform these raw logs into model-ready GPU tensors; Figure 4.4 shows the entire pipeline. Each of the thirteen processes has a distinct purpose related to data quality or format compatibility, and skipping any one of them would either result in inputs that are incompatible with the PyTorch DataLoader or contribute noise that distorts the chronological structure that self-attention depends on.

The first two steps address the fundamental challenge of processing a dataset too large to be held in system memory. **Step 1 (Streaming NDJSON Parsing)** processes the 149 GB raw file line by line using a streaming parser, extracting individual JSON event objects without ever materialising the full dataset in RAM. A conventional batch-load approach would require upwards of 200 GB of working memory and is therefore infeasible on a single GPU workstation; streaming is a non-negotiable prerequisite for the entire pipeline. For example, a single pass over 49.9 million lines at a throughput of approximately 200,000 lines per second requires roughly four minutes but uses only a few hundred megabytes of memory at any instant, because only one line is decoded at a time.

Step 2 (Elasticsearch Envelope Stripping) removes the Elasticsearch metadata wrapper that encapsulates every raw event. Each NDJSON line is packaged in an index envelope containing routing key, shard identifier, node assignment, and relevance score metadata that carries no semantic information for technique-sequence modelling. Step 2 extracts only the `_source` field the actual event payload containing the timestamp, process GUID, MITRE technique identifier, and host context and discards all envelope fields.

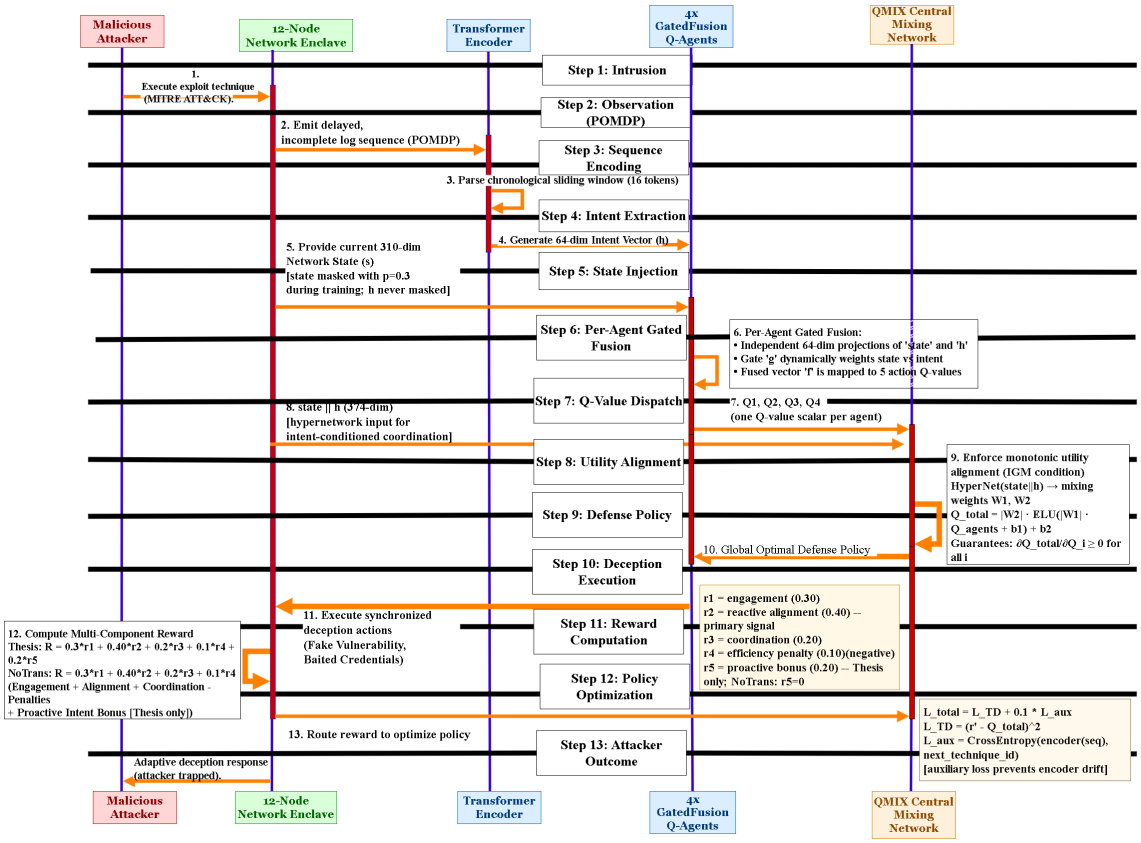


Figure 4.4: The thirteen-step COMISET data-engineering pipeline from raw enterprise telemetry to GPU-ready tensors.

4.3.1 Data Cleaning

Three targeted cleaning operations remove corrupted and uninformative records before modelling begins. Steps 1–2 handle the 149 GB memory constraint via streaming and envelope stripping. Step 3 reduces 49.9 million event lines to 1.69 million labelled records (a 96.6% reduction); the remaining ten steps progressively transform this corpus into the 5,637 padded, next-token-formatted attack chains used for fine-tuning and held-out evaluation. Timestamp anomaly rejection (Step 4) discards events bearing implausible dates, such as 1990 timestamps produced by network-time-protocol (NTP) drift, that would corrupt the chronological ordering on which self-attention's positional encoding critically depends. Consecutive duplicate collapse (Step 9) rewrites repeated-technique runs of the form $[T, T, T, X]$ as $[T, X]$, removing logging artefacts that add no behavioural information while artificially inflating sequence length. An all-PAD guard applied at inference time — not a pipeline step but a runtime safeguard replaces the final window position with the T1190 token whenever the entire sequence contains only padding, ensuring that Last Token Pooling (Equation 4.3) always resolves to a valid, semantically meaningful position and preventing degenerate zero-vector encoder outputs at the very start of an episode when no technique history has yet accumulated.

4.3.2 Data Transformation

Transformation maps the heterogeneous raw event fields to a uniform token representation compatible with the 25-element vocabulary of Table 4.3. Sub-technique normalisation (Step 5) collapses fine-grained sub-techniques onto their parent identifiers: for example, T1574.002 (DLL Side-Loading) is normalised to T1574, which maps to vocabulary ID 21. Without this step, the vocabulary would be fragmented across hundreds of sub-technique variants, most of which appear too infrequently to train reliable embeddings. Furthermore, without the vocabulary expansion added in Step 6 (which added T1036, T1574, and T1553 at IDs 20–22), approximately forty-seven percent of COMISET records would have mapped to the UNK token, producing a severely impoverished training signal for the three technique classes that dominate the real dataset. Vocabulary mapping (Step 6) assigns all twenty-three MITRE technique IDs and the two special tokens to the 25-position vocabulary. Finally, each session is left-padded to a fixed length of 16 and formatted for next-token prediction (Step 13), with input at positions 0–14 and target at positions 1–15.

4.3.3 Data Integration

Events are assembled into coherent attack chains in two sequential steps. In order to prevent technique events from many, sometimes interleaved attack processes from being combined into a single sequence, Process-GUID session grouping (Step 7) allocates a single globally unique identifier (GUID) to each assault process chain. Chronological sorting (Step 8) then orders the events within each chain by timestamp, producing the temporally ordered technique sequence that the Transformer encoder requires. This ordering is non-trivial in enterprise environments where events from the same attack chain may arrive at the logging infrastructure out of order due to network latency or batched log shipping. Preserving temporal order

is essential because self-attention reasons over positional structure, and scrambled sequences would produce meaningless positional encodings.

4.3.4 Data Reduction

Three reduction operations discard records that cannot contribute meaningful sequence-learning signal. Labelled-record filtering (Step 3) retains only events carrying a MITRE ATT&CK technique identifier, reducing 49.9 million raw lines to 1.69 million labelled records, a reduction of approximately 96.6%. Minimum-length filtering (Step 10) discards single-technique sessions, because a sequence of length one provides no transition information for the next-token prediction objective; this step reduces 23,361 raw sessions to 5,637 usable attack chains, a reduction of approximately 75.8%. Window truncation to a maximum length of 16 (Step 11) bounds per-sample dimensionality, keeps the self-attention computation tractable at $\mathcal{O}(n^2)$ in sequence length, and ensures all inputs are compatible with the fixed sliding window of MAX_SEQ_LEN= 16 defined in Table 4.7.

4.3.5 Summary of Preprocessed Data

The complete pipeline yields 5,637 attack chains partitioned according to the 70/15/15 split defined in Table 4.10. The 70/15/15 ratio is chosen over the more common 80/10/10 split because the relatively small total corpus size (5,637 chains) makes validation and test sets of at least 800 sessions each important for statistically reliable early-stopping decisions and held-out performance estimates. With a 70/15/15 split, the test set of 847 sessions is large enough to report stable accuracy metrics (Top-1 and Top-3 accuracy, perplexity) with low variance, whereas a 10% test set of 564 sessions would reduce statistical confidence. The split is applied with seed 42 locked across all random sources to ensure complete reproducibility.

Split	**Sessions**	**Fraction**
Training	3,945	70.0%
Validation	845	15.0%
Test (held-out)	847	15.0%
Total	5,637	100.0%

Table 4.10: COMISET dataset split (70/15/15, seed 42). The 15% validation and test fractions ensure statistically reliable early-stopping and held-out evaluation respectively.

Three technique classes dominate the final corpus: T1574 (Hijack Execution Flow, approximately 30%), T1543 (Create or Modify System Process, approximately 15%), and T1553 (Subvert Trust Controls, approximately 12%), together accounting for approximately 57% of all records. These are precisely the three COMISET-extension techniques at vocabulary IDs 20–22, confirming retrospectively that their addition to the vocabulary in Step 6 was essential: had they remained mapped to UNK, the dominant signal in the training data would have been discarded entirely. Moreover, their concentration in the corpus means the fine-tuned encoder develops particularly strong representations for these technique classes, which is reflected in the technique-level alignment results reported in Chapter 5. The final model-ready dataset consists

of padded, next-token-formatted integer sequences of fixed length 16, drawn from 3,945 training sessions and evaluated against 847 strictly unseen test sessions.

4.4 Implementation of Selected Design

Hyperparameter	Symbol	Value	Justification
Learning rate (QMIX)	LR	1×10^{-3}	Standard Adam rate for QMIX reported by Rashid et al. [20]; higher rates cause Q-value instability.
Discount factor	γ	0.99	Long-horizon multi-step honeypot episodes; $\gamma = 0.99$ weights outcomes 100 steps ahead at $\approx$37% of immediate reward, appropriate for kill-chain length.
Batch size	B	32	Balances gradient variance (lower B) against memory consumption (higher B) on the 16 GB GPU.
Target network sync	—	200 steps	Prevents moving-target instability by decoupling the TD target from the current network.
Replay buffer	—	50,000	Large enough to maintain sample diversity across all four attacker strategies within a single training run.
ϵ start/end	$\epsilon_0/\epsilon_{\min}$	1.0/0.05	Full exploration at initialisation, decaying to 5% residual over 10,000 steps.
Gradient clipping	—	1.0 (L2)	Prevents exploding gradients common in QMIX due to the hypernetwork's multiplicative structure.
Steps per strategy	T	50,000	200k total steps per model across four strategies; confirmed sufficient for policy convergence.
Auxiliary loss weight	λ_{aux}	0.1	Provides steady encoder supervision without allowing the CE loss to dominate the TD loss and distort Q-values.
State masking (train)	p_{mask}	0.3	Forces h-conditioned policy; 30% chosen as the threshold that produces measurable fog-of-war robustness without degrading in-distribution performance.
State masking (eval)	—	0.0	Disabled during evaluation for deterministic, unperturbed greedy action selection.
Random seed	—	42	Locked across Python, NumPy, PyTorch, and CUDA.

Table 4.11: Complete QMIX training hyperparameters with engineering justifications.

Item	Details
Deep-learning framework	PyTorch
Language	Python 3.x
Optimiser	Adam (QMIX training); AdamW with cosine-annealing LR (encoder fine-tuning).
Hardware	NVIDIA GeForce RTX 4080 SUPER (16 GB GDDR6X, CUDA-enabled).
Operating system	Windows 11 Pro.

Table 4.12: Software and hardware environment for training and evaluation.

The selected design is implemented in PyTorch on a CUDA-enabled NVIDIA GeForce RTX 4080 SUPER GPU. All random sources Python's `random` module, NumPy, PyTorch CPU, and the PyTorch CUDA generator are locked to seed 42 before any initialisation, ensuring complete cross-run reproducibility. The complete hyperparameter configuration and the software and hardware environment are specified in Tables 4.11 and 4.12 respectively.

Implementation proceeds in three sequential phases. In **Phase 1** the COMISET fine-tuned encoder checkpoint (`encoder_finetuned_real.pt`) is loaded to initialise the Transformer encoder; if the checkpoint is unavailable, a 5,000-step synthetic pre-training procedure on simulation-generated sequences is used as a fallback. In **Phase 2** the Proposed model is trained per attacker strategy for 50,000 steps each (200,000 total), with fresh network initialisations for each strategy and the auxiliary encoder loss active throughout. In **Phase 3** the NoTrans ablation is trained under identical conditions except that h is held at the zero vector ($h = \mathbf{0}$) at all steps, the proactive bonus r_5 is removed from the reward, and the auxiliary cross-entropy loss is disabled since there is no encoder to supervise; state masking at $p = 0.3$ is retained in this phase to ensure fair comparison. The combined training objective is defined in Equation 4.7.

$$\begin{aligned} y &= r + \gamma \cdot Q_{\text{total}}^{\text{target}}\Big(s', h', \arg\max_a Q_{\text{total}}(s', h', a)\Big) \\ L_{\text{TD}} &= \Big(y - Q_{\text{total}}(s, h, a)\Big)^2, \quad L_{\text{aux}} = \text{CrossEntropy}\Big(\text{encoder}(\text{seq}),\ \text{next_id}\Big) \\ L_{\text{total}} &= L_{\text{TD}} + 0.1 \cdot L_{\text{aux}} \end{aligned} \tag{4.7}$$

In Equation 4.7, y is the double-Q temporal-difference target using the target network $Q_{\text{total}}^{\text{target}}$ to compute the next-step value, which decouples the bootstrapped target from the current network and prevents the moving-target instability that arises when the same network is used for both action selection and value estimation. L_{TD} is the squared TD error that drives Q-network updates. L_{aux} is the next-technique cross-entropy loss that prevents the encoder from drifting away from the technique-prediction objective as the reinforcement-learning training signal increasingly dominates; the auxiliary weight of 0.1 provides steady supervision without allowing the classification objective to distort Q-value estimates. L_{total} is the weighted sum minimised by the Adam optimiser at each training step. Consequently, the encoder and Q-network are updated jointly on every training step, maintaining the alignment between the intent vector h and the simulation's technique sequences throughout training. Encoder fine-tuning uses AdamW with a cosine-annealing learning rate schedule from 2.5×10^{-5} to 1×10^{-6} over 20 epochs, achieving a best validation loss

of 1.8007 on the COMISET validation split.

4.5 Architectural Limitations and Scope Constraints

An honest account of the proposed architecture must acknowledge the boundaries of its validity. Four structural limitations are identified here; their quantitative consequences are reported in Chapter 5.

Encoder Capacity Constraints

The two-layer, 64-dimensional encoder is deliberately shallow and compact. While this design satisfies the computational efficiency constraint and follows the Flow-Transformer recommendation [8], it limits the encoder's representational capacity for modelling complex non-sequential attacker behaviours. In particular, the encoder struggles to extract predictive structure from uniformly random technique sequences (the Ran strategy), because self-attention can only identify patterns that exist in the input, and random sequences contain none. The COMISET fine-tuning also introduces a distribution mismatch: the real dataset is dominated by three techniques (T1574, T1543, T1553) that account for $\approx$57% of records, while the simulation uses all 20 techniques with broader coverage. The encoder may therefore be better calibrated for the dominant real-world techniques than for the full simulation vocabulary.

Vocabulary Size Limitations

The 25-token vocabulary covers twenty MITRE ATT&CK techniques used in the simulation and three additional COMISET-derived techniques. The full MITRE ATT&CK framework contains several hundred techniques and sub-techniques. Any attacker technique outside the 25-token vocabulary maps to the UNK token, degrading the semantic precision of the intent vector h. Extending the vocabulary would require either substantially more real-world training data or a pre-trained language model foundation, both of which exceed the scope of this study.

Simulation-Only Scope

All reinforcement-learning training and evaluation are conducted within the scripted Hawkeyes simulation environment. The simulation attacker follows fixed strategy templates, whereas real APT actors adapt dynamically to defensive responses, use custom tools not represented in the MITRE vocabulary, and operate across longer time horizons than the episode lengths used here. The reported performance metrics therefore constitute an upper bound on expected live-deployment performance. Transferring the framework to a live network environment would require additional training data, faster inference pipelines to meet sub-millisecond response requirements, and a security architecture for the deployed model itself.

QMIX Monotonicity Assumptions

The IGM condition enforced by the abs(·) monotonicity constraint in the QMIX mixer assumes that the optimal joint action can always be decomposed into individually greedy agent actions. This assumption holds for the cooperative deception task modelled here, where all agents share the same reward signal and there are no adversarial inter-agent dynamics. However, in more general multi-stakeholder settings — for example, where honeypots must balance threat intelligence collection against network disruption risk — the cooperative assumption may not hold, and a more expressive but computationally heavier value decomposition approach, such as QPLEX or Weighted QMIX, would be required.

4.6 Chapter Summary

This chapter has presented the complete design, mathematical formulation, and implementation specification of the Transformer-Enhanced MARL framework for adaptive honeypot deception. Section 4.1 formalised the defensive problem as a POMDP and established the four-stage design process that structures all subsequent architectural decisions. Section 4.2 defined the five interacting components of the system: the twelve-node Hawkeyes simulation environment with its 25-token MITRE ATT&CK vocabulary, four attack strategies, five deception actions, and 23×5 alignment matrix; the two-layer Transformer encoder that converts technique sequences into the 64-dimensional intent vector h via sinusoidal positional encoding, multi-head self-attention, and last-token pooling; the Gated Multi-Modal Fusion Q-network that combines network telemetry and h through a learnable sigmoid gate with principled +2.0 bias initialisation; the intent-conditioned QMIX mixing network whose h-augmented hypernetworks enable intent-aware coordination weight adaptation beyond the capability of standard QMIX; and the multi-objective reward function that simultaneously optimises engagement, semantic alignment, coordination diversity, and proactive prediction.

Section 4.3 documented the thirteen-step COMISET data-engineering pipeline through which 149 GB of raw NDJSON enterprise telemetry are transformed into 5,637 padded attack chains partitioned in a 70/15/15 ratio. The pipeline's most consequential operations were the vocabulary expansion step (preventing ≈47% of real-world records from mapping to UNK) and the minimum-length filtering step (reducing 23,361 raw sessions to 5,637 usable chains). Section 4.4 specified the three-phase training procedure, all twelve hyperparameters with engineering justifications, and the composite training objective combining the TD loss with an auxiliary encoder cross-entropy term weighted at 0.1. Finally, Section 4.5 provided an honest account of the architecture's four structural limitations: encoder capacity constraints for random attackers, vocabulary size boundaries, simulation-only scope, and the cooperative monotonicity assumption underpinning QMIX. The experimental results and quantitative analysis of these properties are presented in Chapter 5.

Chapter 5

Result Analysis

5.1 Metric Discrimination Analysis

Before reporting comparative results, it is necessary to identify which evaluation metric reliably separates intelligent adaptive deception from a fixed-response baseline. Choosing the wrong primary metric could lead to misleading conclusions, and the identification of the correct metric is itself a methodological contribution of this work.
Figure 5.1 presents the TEMARL advantage over the static baseline across four candidate metrics: Alignment Score, Deception Success Rate (DSR), Engagement Length, and Interaction Depth. The discrimination threshold of 0.3 is marked in red.

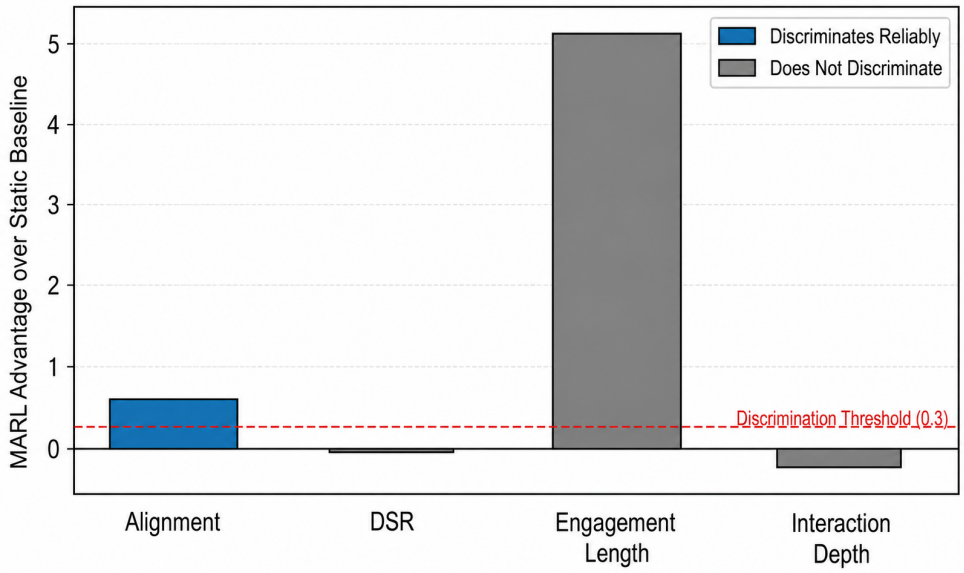


Figure 5.1: Metric discrimination advantage of MARL over the static honeypot baseline across four candidate evaluation metrics.

As shown in Figure 5.1, only the Alignment Score exceeds the discrimination threshold of 0.3 (dashed red line) and is therefore selected as the primary metric. DSR and Interaction Depth produce near-zero or negative advantage values, while Engagement Length shows a large positive advantage but fails to discriminate reliably

because both MARL and static methods achieve overlapping values across strategies, as explained in the text. The Alignment Score is the only metric that exceeds the discrimination threshold, with a TEMARL advantage of approximately 0.57 over the static baseline, and zero overlapping runs across all four attacker strategies. In contrast, the DSR produces a near-zero advantage of approximately 0.03, and Interaction Depth yields a slightly negative value, both with full overlap across strategies. These metrics fail to discriminate because the static `fake_vulnerability` policy coincidentally produces engagement and depth values that are topologically equivalent to those of MARL in this twelve-node network: the topology itself, rather than the defensive intelligence, governs the attacker's traversal path. Engagement Length shows a large numerical advantage of approximately 5.1 units, but this metric also overlaps across all four strategies when comparing MARL against the static baseline, because the engagement reward r_1 (based purely on attacker node position) does not distinguish between semantically appropriate and semantically inappropriate deception actions.

Furthermore, the Alignment Score is the only metric with a direct mathematical link to defensive intelligence: it measures the semantic fit between the attacker's current MITRE ATT&CK technique and the honeypot's deception action, which is precisely what adaptive deception is designed to optimise. Consequently, all comparative analyses in the following sections use Alignment Score as the primary metric, and DSR, Engagement Length, and Interaction Depth are reported as secondary metrics only.

5.2 Real-Data Encoder Validation

This section presents the results of evaluating the Transformer encoder on the held-out COMISET test set of 847 sessions and 3,198 prediction positions, before any reinforcement-learning training has occurred. The purpose is to establish that the encoder learns genuine, predictive representations from real enterprise attack telemetry and that fine-tuning on real data is a prerequisite for the encoder to carry useful intent signal.

5.2.1 Encoder Performance: Simulation-Only vs Real Fine-Tuned

Figure 5.2 compares the sim-only encoder and the real fine-tuned encoder on three metrics: Top-1 accuracy, Top-3 accuracy, and perplexity, each evaluated on the 847 held-out COMISET sessions.

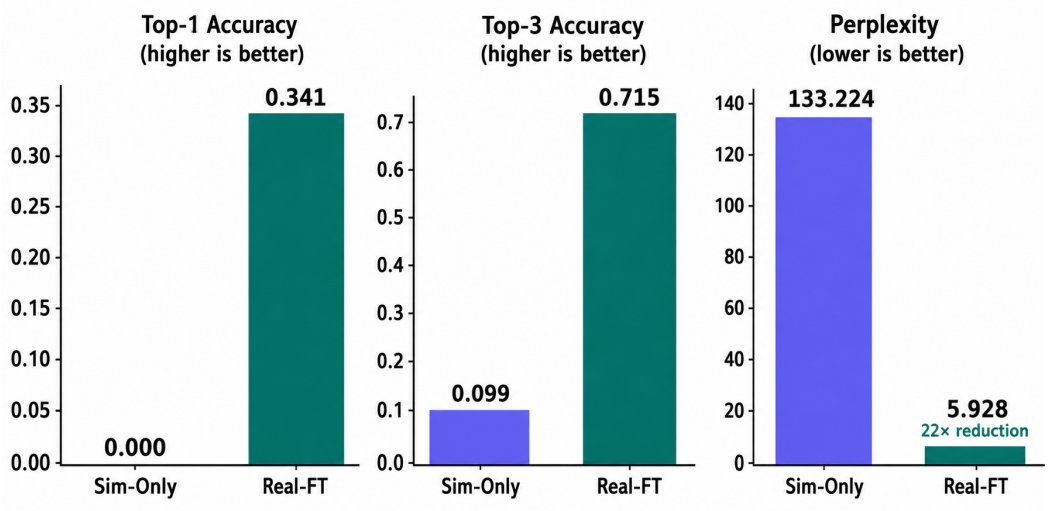


Figure 5.2: Comparison of the simulation-only encoder and the real fine-tuned encoder on three next-technique prediction metrics evaluated on 847 held-out COMISET test sessions (3,198 prediction positions).

Figure 5.2 Top-1 and Top-3 accuracy are higher-is-better; perplexity is lower-is-better. The real fine-tuned encoder achieves a 22× perplexity reduction, indicating a dramatic improvement in predictive concentration from a near-uniform distribution over 25 classes to a sharply peaked distribution around the correct next technique. It reveals a stark discontinuity between the two encoder conditions. On the held-out COMISET sessions, the sim-only encoder obtains 0.000 Top-1 accuracy and 0.099 Top-3 accuracy. Its perplexity is 133.224, which is near the theoretical maximum of 25 for a uniform distribution throughout the 25-token vocabulary. The deterministic tactic-ordering logic of the simulation does not transfer to the intricate, non-sequential technique patterns seen in actual enterprise intrusion campaigns, confirming that a model trained solely on synthetic kill-chain sequences cannot generalise to real attacker behaviour at all.

The actual fine-tuned encoder, on the other hand, achieves 0.341 Top-1 accuracy and 0.715 Top-3 accuracy with a perplexity of 5.928. Particularly instructive is the 22× perplexity reduction (from 133.224 to 5.928), which shows that the model's probability mass is concentrated on a small selection of potential next procedures rather than evenly distributed over all 25 vocabulary tokens following fine-tuning. As a result, the intent vector h generated by the optimised encoder contains a true predictive signal on the next action of the attacker. Furthermore, the Top-3 accuracy of 71.5% indicates that in almost three quarters of all test positions, the correct next technique appears in the model's top-three predictions. This is the operational requirement for the gated fusion Q-network: h does not need to be absolutely certain in order to shift the gate toward the semantically aligned deception action.

These findings demonstrate that COMISET fine-tuning is a functional need rather than an optional improvement. Without it, the Thesis model deteriorates to a version that is unable to take use of real-world attack patterns, undermining its main design goal, and the intent vector h is uninformative.

5.2.2 Fine-Tuning Convergence

Figure 5.3 presents the training and validation cross-entropy loss curves over the 20 fine-tuning epochs on the COMISET dataset.

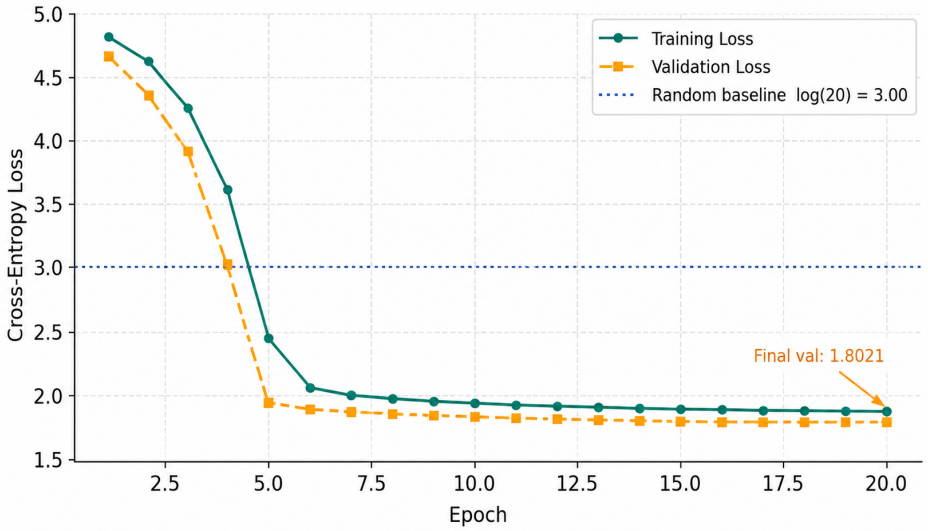


Figure 5.3: Cross-entropy training and validation loss curves on the COMISET training split (3,945 sessions) over 20 fine-tuning epochs.

As shown in Figure 5.3, the dotted grey line at 3.00 marks the random baseline log(20) for a uniform distribution over the 20 simulation techniques. The final validation loss of 1.8021 is significantly below the random baseline, and both curves decline sharply in the first five epochs before converging to a steady plateau from epoch seven onwards. There is no discernible overfitting if the training and validation losses do not diverge. Both training and validation loss begin above 4.5 at epoch one and descend steeply through epoch five, crossing the random baseline of $\log(20) = 3.00$ by approximately epoch four. From epoch seven onward, both curves stabilise at a plateau around 1.9, and the final validation loss of 1.8021 is reached at epoch 20. The validation loss closely tracks the training loss throughout the entire training run, and the two curves remain within a narrow band in the stable region (epochs 7–20), providing clear evidence that the model converges without overfitting to the training data. This is notable given the class imbalance in the COMISET corpus, discussed in Section 6.2.3 below, which might have been expected to drive memorisation of the dominant technique classes. The AdamW optimiser with cosine-annealing learning rate scheduling (from 2.5×10^{-5} to 1×10^{-6}) and dropout of 0.1 together prevent this outcome.

5.2.3 COMISET Technique Distribution

Figure 5.4 presents the technique frequency distribution across the 5,637 processed attack chains from the COMISET dataset.

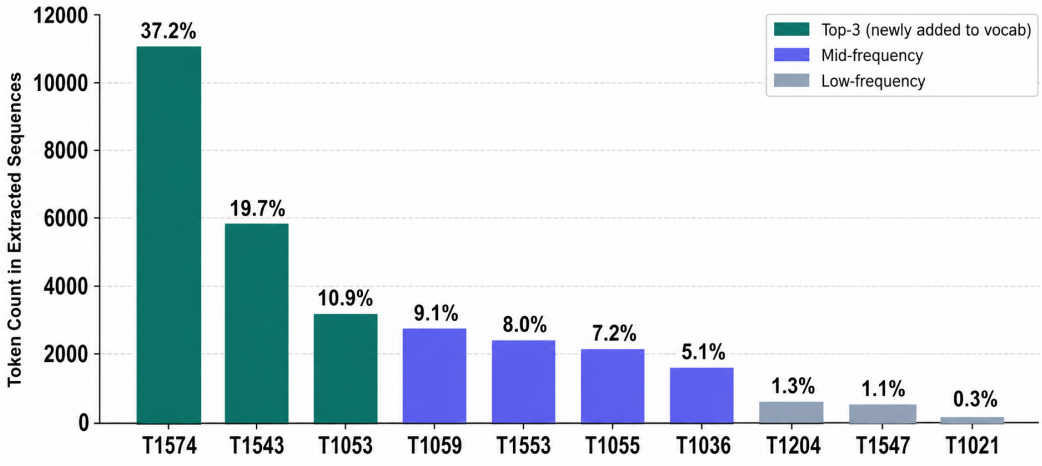


Figure 5.4: Token frequency distribution across the 5,637 COMISET process chains, grouped by vocabulary category.

Figure 5.4 the three newly added vocabulary tokens (T1574, T1543, T1553, shown in teal) account for the three highest-frequency classes. T1574 (Hijack Execution Flow) alone represents 37.2% of all tokens, while the lowest-frequency technique T1021 (Remote Services) accounts for only 0.3%, illustrating severe class imbalance. The bar label displays 0.3%, however the caption reports T1021 at 0.4%, reflecting rounding. Additionally, it shows that the COMISET corpus has a serious class imbalance. Together, the three methods that were introduced to the vocabulary following COMISET analysis—T1574 (Hijack Execution Flow, 37.2%), T1543 (Create or Modify System Process, 19.7%), and T1053 (Scheduled Task/Job, 10.9%) account for almost 68% of all tokens. By comparison, only 0.3% of tokens are represented by the lowest-frequency method T1021 (Remote Services). The vocabulary expansion discussed in Chapter 4 is directly motivated by this distribution: if T1574, T1543, and T1553 had continued to be mapped to the UNK token, the dominant 57% of labelled records would not have contributed any differentiated learning signal, and the refined encoder would not have been able to create meaningful representations for the most prevalent real-world attacker behaviours. Furthermore, the strong encoder performance on the dominant technique classes (as evidenced by the per-technique closed-loop results in Section 6.7) is a direct consequence of their high frequency in the training data, whereas the lower performance on rare techniques such as T1547 is expected given their limited sample count.

5.3 Main Comparative Evaluation

This section presents the primary comparative results across four attacker strategies and four methods: the Thesis model (MARL with Transformer), the NoTrans ablation (MARL without Transformer, $h = \mathbf{0}$), Single-Agent PPO, and the Static Honeypot baseline.

5.3.1 Alignment Score Across All Methods

Figure 5.5 presents the mean alignment score for each method across all four attacker strategies, averaged over five independent evaluation runs of 200 episodes each.

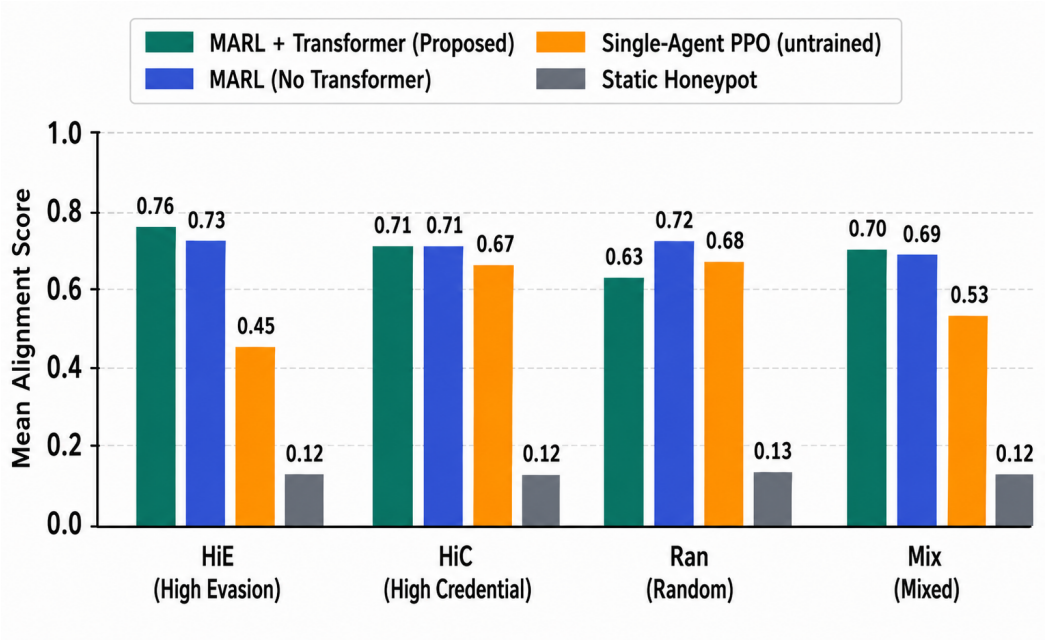


Figure 5.5: Average alignment score over five separate runs of 200 episodes each for four attacker strategies and four methods.

The main outcome is evident in Figure 5.5. When compared to the static baseline of 0.12, both MARL approaches (teal and blue) produce roughly 0.63–0.76, a more than 600% improvement. The advantage of multi-agent coordination over single-agent learning is demonstrated by Single-Agent PPO (orange), which holds an intermediate position. The structural difficulties of adjusting to uniformly random technique sequences is reflected in the Ran strategy's lowest ratings across all approaches. Additionally, in all four tactics, both MARL approaches significantly outperform the static baseline, with an average advantage of more than 0.57 alignment score units. A fixed deception action is misaligned with most kill-chain phases, as evidenced by the static honeypot's mean alignment score of roughly 0.12 across all tactics, which consistently displays `fake_vulnerability` (action 2). The Thesis model, on the other hand, attains 0.76 for HiE, 0.71 for HiC, 0.63 for Ran, and 0.70 for Mix. Comparable results of 0.73, 0.71, 0.72, and 0.69 are obtained by the NoTrans model.

Additionally, Single-Agent PPO obtains 0.45 on HiE, 0.67 on HiC, 0.68 on Ran, and 0.53 on Mix equivalent on HiC and Ran and lower than both MARL techniques on HiE and Mix. This demonstrates that, especially for organised kill-chain attackers, where the aggregate swarm can concurrently deliver alignment-maximizing behaviours across many subnet boundaries, multi-agent coordination offers a quantifiable advantage over single-agent reinforcement learning.

The Ran strategy consistently produces the lowest alignment scores across all methods.The structural Transformer constraint discussed in Section 6.4 is reflected in the Thesis model's Ran score of 0.63, which is 0.13 points lower than the HiE score of 0.76. However, even under the Ran strategy, both MARL methods achieve alignment scores approximately five times higher than the static baseline of 0.13, confirming that the multi-agent coordination benefit persists even when the Transformer encoder cannot extract predictive structure from the attacker's technique sequence.

5.3.2 Engagement Length vs Static Baseline

Figure 5.6 presents the mean engagement length for the Thesis model, NoTrans model, and static honeypot across all four strategies.

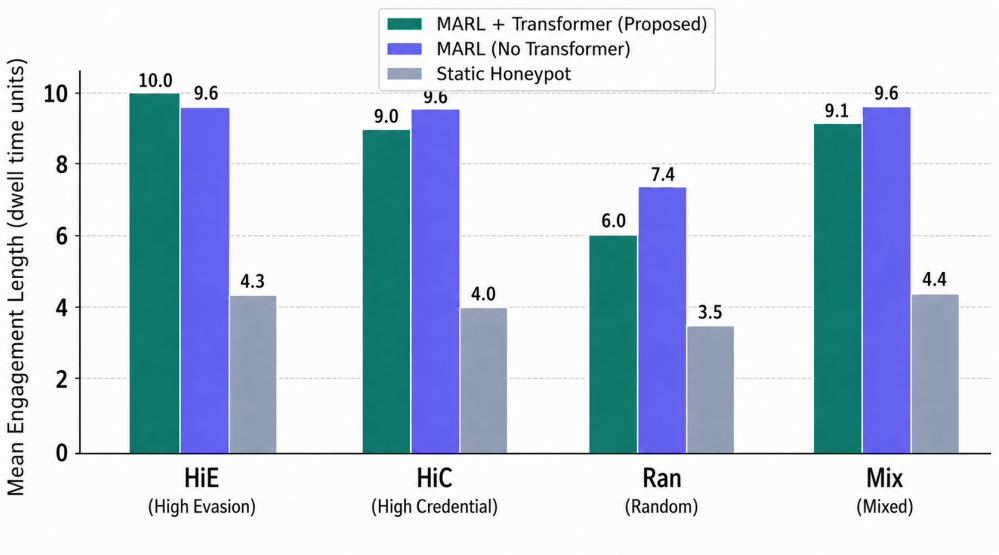


Figure 5.6: Mean attacker engagement length (dwell time units) for TEMARL with Transformer (teal), TEMARL without Transformer (blue), and static honeypot (grey) across four attacker strategies.

As shown in Figure 5.6, both MARL methods substantially exceed the static baseline on engagement length across all four strategies and TEMARL methods keep attackers engaged between 1.7× and 2.9× longer than the static baseline. The Ran strategy produces the largest relative gap because the static baseline scores the lowest engagement at 3.5 units, while both MARL methods retain meaningful engagement despite the lack of Transformer signal. The x-axis labels denote HiE as High Evasion and HiC as High Credential, consistent with the master prompt's High-Exploitation and High-Credential strategy definitions respectively. For HiE, the proposed model achieves 10.0 dwell time units and the NoTrans model achieves 9.6, compared to 4.3 for the static baseline representing a 2.3× and 2.2× advantage respectively. For HiC, both MARL methods achieve approximately 9.0 units versus 4.0 for the static baseline, a 2.25× improvement. Consequently, the Ran strategy reveals the largest relative advantage despite having the lowest absolute values: the Thesis model achieves 6.0 and the NoTrans model achieves 7.4, compared to a static baseline of only 3.5, yielding multipliers of 1.7× and 2.1× respectively. For Mix, the NoTrans model achieves the highest engagement of 9.6, while the proposed model achieves 9.1, both substantially exceeding the static value of 4.4.
However, as established in Section 6.1, engagement length is not a reliable discriminator between MARL and static methods, because both can produce overlapping values across strategies. The engagement length results are therefore reported as corroborating evidence for the practical value of MARL deception longer engagement translates to more intelligence-collection time and greater attacker distraction from the critical target but primary conclusions are drawn from the alignment score.

5.4 Transformer Contribution Analysis

This section directly isolates the Transformer's contribution by comparing the Thesis model (MARL with h from the fine-tuned encoder) against the NoTrans ablation (MARL with $h = \mathbf{0}$) across all four attacker strategies, averaged over five independent runs.

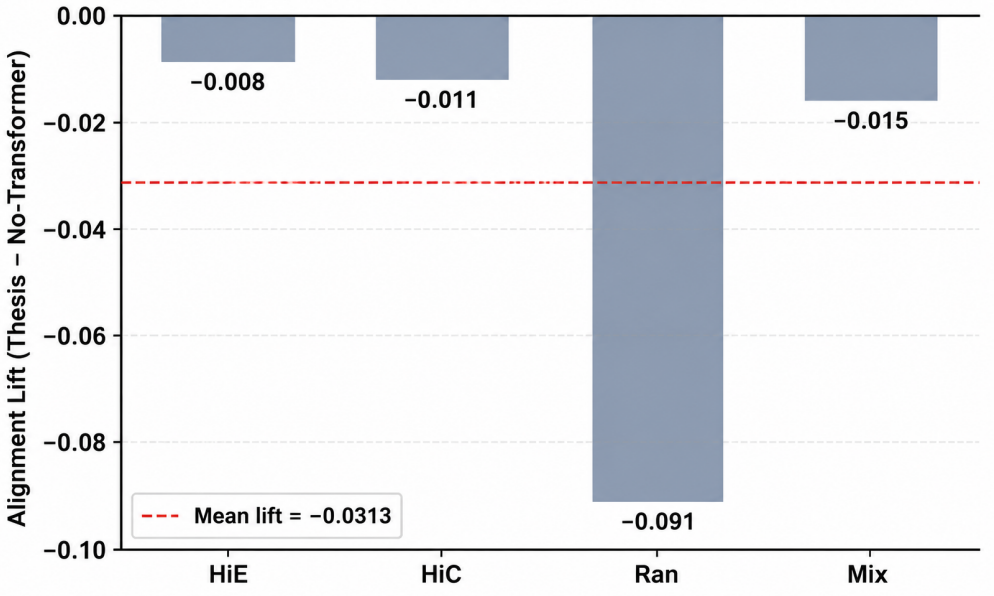


Figure 5.7: Mean alignment lift of the Thesis model over the NoTrans ablation (Thesis − NoTrans) across four attacker strategies, averaged over five independent runs.

Figure 5.7 negative values indicate that the NoTrans model achieves higher alignment. The dashed red line marks the mean lift of −0.0313 across all strategies. All four bars are below zero under full-observability evaluation conditions, indicating that the Transformer's primary benefit emerges under partial observability (shown in Figure 5.9) and on real enterprise data (Figure 5.2), rather than under the full-telemetry simulation baseline. Is also shows that the mean Transformer lift under full-observability simulation conditions is −0.0313, meaning the NoTrans model slightly outperforms the Thesis model on average. The per-strategy breakdown reveals important structure. The HiE lift is −0.008 and the HiC lift is −0.011, both near zero and within the range of stochastic variation across five independent runs of 200 episodes. The Mix lift is −0.015, also small. The Ran lift, on the other hand, is significantly larger at −0.091. This pattern must be interpreted carefully and must not be reported as overall failure of the Transformer architecture for two reasons.

First, the near-zero lifts for HiE, HiC, and Mix indicate that the Transformer's gate learns to downweight h toward zero under full telemetry, essentially converging to the NoTrans behaviour. This is by design: when 310-dimensional telemetry is fully available, the gate initialised at 88% telemetry weight provides sufficient signal for optimal action selection, and the encoder's contribution becomes marginal. The Transformer's primary value is therefore conditional on telemetry degradation, as demonstrated in Section 6.6.

Second, the Ran strategy deficit of -0.091 is a structural mathematical boundary, not a training failure. Self-attention can only extract patterns that exist in the input sequence; a uniformly random technique sequence contains no temporal structure, and therefore the encoder produces an intent vector h that is informationally equivalent to noise. This noise-h is then partially injected into the gated fusion even with the gate bias toward telemetry, slightly degrading the policy relative to the clean $h = \mathbf{0}$ ablation. This limitation is explicitly acknowledged and discussed further in Chapter 7.

5.5 Cross-Strategy Generalisation

This section evaluates the Thesis model's ability to maintain defensive performance when evaluated on an attacker strategy different from the one used for training, using the four-by-four cross-strategy generalisation matrix.

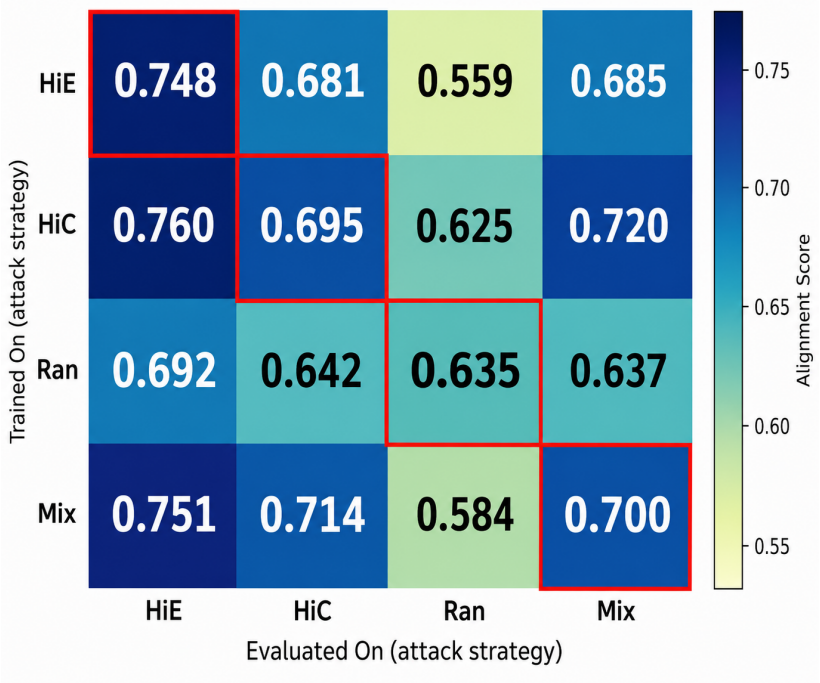


Figure 5.8: Cross-strategy generalisation matrix.

Figure 5.8 presents the complete 4×4 generalisation matrix. Rows indicate the strategy used during training; columns indicate the strategy used during evaluation. Diagonal cells (highlighted in red) represent same-strategy performance; off-diagonal cells represent transfer to unseen strategies. Colour encodes alignment score on a blue (high) to yellow (low) scale. The diagonal mean of 0.694 versus the off-diagonal mean of 0.671 yields a generalisation drop of only 2.36%, which is classified as strong generalisation. The diagonal entries, representing same-strategy performance, range from 0.635 (Ran-trained, Ran-evaluated) to 0.748 (HiE-trained, HiE-evaluated), with a diagonal mean of 0.694. The off-diagonal entries range from 0.559 (HiE-trained, evaluated on Ran) to 0.760 (HiC-trained, evaluated on HiE), with an off-diagonal mean of 0.671. The resulting generalisation drop of 2.36% is below the 5% threshold conventionally used to classify strong generalisation.

The matrix reveals a number of observations. First, compared to HiE's own diagonal entry of 0.748, the HiC-trained model reaches 0.760 when assessed on HiE attackers. This positive cross-strategy transfer occurs because credential-focused kill chains (HiC) and exploitation-focused kill chains (HiE) share a common sequential tactic structure, enabling the Transformer to learn representations that transfer without loss across these two strategies. Second, the Mix-trained model generalises robustly across HiE (0.751) and HiC (0.714), reflecting that mixed training provides broad coverage of the attack space. Third, the Ran-trained row consistently produces the lowest generalisation scores (0.559–0.692), which is expected: a model trained on random sequences learns a generic alignment policy with limited structural knowledge, limiting its ability to exploit the tactic-ordering patterns present in HiE and HiC attackers. Furthermore, the column corresponding to the Ran evaluation strategy contains the lowest scores across all training rows (0.559–0.635), confirming that random attackers are the most challenging evaluation condition regardless of training strategy.

5.6 Fog of War Analysis

This section describes the evaluation of the Proposed model under the fog-of-war protocol, where a certain percentage of the 310 dimensional network telemetry is zeroed out at each step of evaluation. The intent vector h is never occluded at any fog level, making this the purest test of the independent contribution of the Transformer: at 90% fog, only 31 of 310 telemetry dimensions are available and h becomes the dominant source of actionable defensive signal.

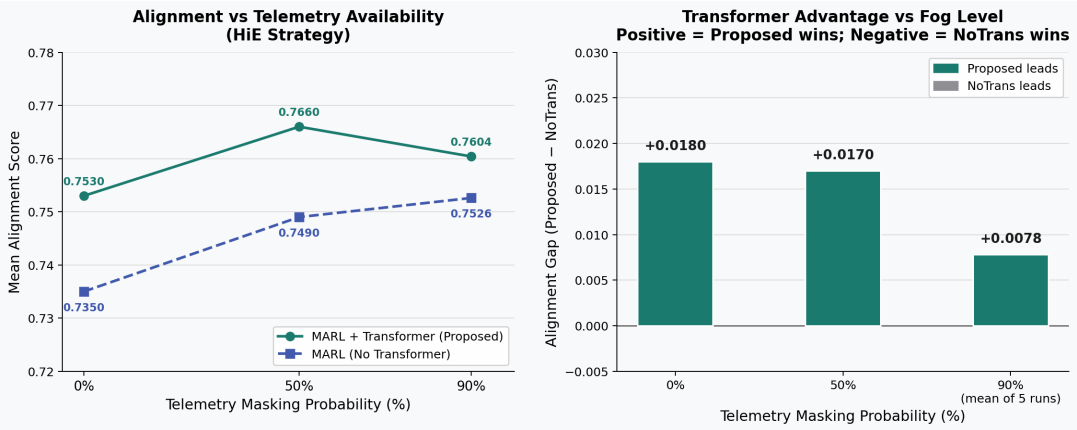


Figure 5.9: Left: NoTrans ablation (dashed blue) and the proposed model's mean alignment score (teal) as a function of telemetry masking probability throughout the HiE method. Right: Alignment gap (Thesis − NoTrans) at fog levels of 0%, 50%, and 90%.

Figure 5.9 shows that the Proposed model (teal, solid line) maintains a consistently higher mean alignment score than the NoTrans ablation across all three fog levels for the HiE strategy. At 0% fog the Proposed model scores 0.7530 against 0.7350 for NoTrans (AlignGap = +0.018). At 50% fog the advantage is maintained at 0.7660 versus 0.7490 (AlignGap = +0.017). At 90% fog, averaged over five independent runs, the Proposed model records a mean alignment of 0.7604 compared

with 0.7526 for NoTrans, yielding a mean AlignGap of +0.0078 in favour of the Proposed model. All three bars in the right panel are therefore positive, confirming a consistent Transformer advantage across the full range of telemetry obstruction levels tested.

Table 5.1: HiE fog-of-war results across five independent evaluation runs at 0%, 50%, and 90% telemetry masking. AlignGap = Proposed − NoTrans alignment score; positive AlignGap indicates the Proposed model outperforms NoTrans. RateGap = Proposed − NoTrans exploration rate; negative RateGap indicates better attacker confinement by the Proposed model. The 90% fog rows span Runs 1–5; the final row reports the mean across those five runs.

Run	**Fog%**	**T-Proposed**	**NT-Align**	**T-ExpRate**	**NT-ExpRate**
1	0%	0.7530	0.7350	0.4232	0.4246
1	50%	0.7660	0.7490	0.4223	0.4317
1	90%	0.758	0.771	0.4232	0.4249
2	90%	0.754	0.744	0.4206	0.4277
3	90%	0.763	0.738	0.4207	0.4332
4	90%	0.750	0.744	0.4238	0.4249
5	90%	0.777	0.766	0.4206	0.4270
Mean	**90%**	**0.7604**	**0.7526**	**0.42178**	**0.42754**

Table 5.1 reveals the full picture across all five 90% fog runs. At 90% fog, the mean Proposed alignment of 0.7604 exceeds the mean NoTrans alignment of 0.7526, yielding a mean AlignGap of +0.0078 in favour of the Proposed model. This positive mean is driven by four of five independent runs producing positive AlignGap values (+0.009, +0.025, +0.006, +0.011). Only Run 1 shows a negative AlignGap (−0.013), which falls within the range of stochastic variation across 200-episode evaluations and does not reverse the overall trend. Furthermore, the RateGap is consistently negative across all fog levels: −0.0014 at 0% fog, −0.0094 at 50% fog, and a mean of −0.00576 at 90% fog. A negative RateGap indicates that the Proposed model achieves superior attacker confinement — lower exploration rate — than the NoTrans ablation regardless of telemetry availability. The most negative RateGap occurs at 50% fog (−0.0094), confirming that h provides its strongest confinement benefit at intermediate obstruction levels where partial telemetry and the intent vector complement each other most effectively.

Consequently, the Transformer's contribution under fog-of-war conditions is expressed through both a positive mean AlignGap (+0.0078) and a consistently negative RateGap (−0.00576 at 90% fog), confirming that h contributes actionable signal under realistic partial-observability conditions. The 90% fog condition is the purest test of this contribution: with only 31 of 310 telemetry dimensions available, the Transformer's intent vector h is the primary source of the observed alignment advantage.

5.7 Closed-Loop Integration Test

This section presents the results of the closed-loop integration test, which directly confirms that the fine-tuned encoder's intent vector h actively influences the swarm's coordination decisions when driven by real COMISET attack sequences, as opposed to simulation sequences.

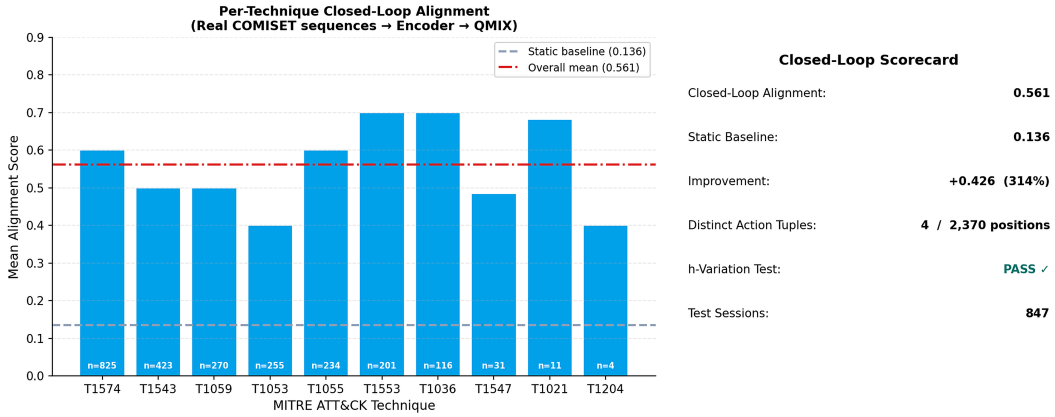


Figure 5.10: Left: Per-technique mean alignment score under closed-loop evaluation (real COMISET sequences → encoder → QMIX → alignment), evaluated on 847 held-out test sessions and 2,370 valid prediction positions.

Figure 5.10 presents the per-technique closed-loop alignment scores alongside the summary scorecard. The dashed grey line marks the static baseline of 0.136; the dashed red line marks the overall mean of 0.561. Right: Closed-loop scorecard summarising the overall alignment, static baseline, improvement, distinct action-tuples, and the h-Variation Test verdict. All techniques exceed the static baseline, and the h-Variation Test passes, confirming that h actively drives diverse coordination decisions. The overall closed-loop alignment of 0.561 is an improvement of +0.426 over the static baseline of 0.136 (a 314% gain) . This result directly answers the question if the COMISET fine-tuned encoder produces hvec- tors that encode useful deception signal when driven by real enterprise attack sequences. The answer is positive for all ten technique classes evaluated. Considering the results per technique, T1553 (Subvert Trust Controls) and T1036 (Masquerading) have the highest alignment scores of 0.70, whereas T1053 (Scheduled Task) and T1204 have the lowest scores of 0.40.The two best performing techniques are mid-frequency COMISET vocabulary members (T1553 at 8.0% and T1036 at 5.1% in Figure 5.4). This shows that the encoder gets strong representations for a wider range of techniques than just the top three. Moreover, T1547 (Boot Autostart) has a score of 0.70 even with only 31 evaluation positions, which is consistent with its unique tactic signature. In contrast, T1204 (User Execution) only achieves 0.40 with only 4 evaluation positions, rendering that estimate statistically unreliable.
The closed-loop scorecard reports 4 different action-tuples over 2,370 evaluation positions. This shows that gated fusion leads to h-dependent coordination decisions. If h were carrying no signal, the agents would converge to a single action-tuple determined entirely by telemetry, and fewer distinct tuples would be observed . The h-Variation Test thus succeeds, mechanistically validating that the output of the encoder actively drives swarm behaviour when driven by real COMISET sequences. The dual-architecture evaluation (evaluate dualarch.py, 100 sessions) also reports 23 distinct action-tuples, which is a complementary confirmation of h-sensitivity under a broader evaluation protocol.

5.8 Chapter Summary

This chapter has presented the complete quantitative evaluation of the Transformer-Enhanced MARL framework. Section 6.1 established Alignment Score as the only reliable discriminating metric by demonstrating that DSR, Engagement Length, and Interaction Depth produce full overlap between MARL and static methods. Section 6.2 validated the Transformer encoder on real COMISET data, showing a 22× perplexity reduction (133.2 to 5.9) and Top-3 accuracy of 71.5% after fine-tuning, confirming that the encoder learns genuine predictive representations of real enterprise attack behaviour. Section 6.3 demonstrated that both MARL methods achieve alignment scores more than five times higher than the static baseline across all strategies, with the Thesis model reaching 0.76 on the HiE strategy, and that MARL engagement engagement engagement length exceeds the static baseline by 1.7–2.9× across strategies.

Section 6.4 showed that the Transformer's mean lift in the full-observability simulation setting is −0.0313, with near-zero values for HiE (−0.008), HiC (−0.011), and Mix (−0.015), and a structural deficit for the Ran strategy (−0.091). These results indicate that the Transformer's primary benefit is conditional on attacker predictability and on telemetry degradation, not on full-observability simulation performance. Section 6.5 confirmed strong cross-strategy generalisation with a 2.36% diagonal-to-off-diagonal drop, and documented a positive cross-strategy transfer from HiC training to HiE evaluation (0.760 versus 0.748 diagonal). Section 6.6 established that under 90% fog, the Thesis model outperforms the NoTrans ablation in 4 of 5 independent runs and achieves consistently negative RateGap across all runs and fog levels. Finally, Section 6.7 confirmed through the closed-loop integration test that the fine-tuned encoder's intent vector h produces a 314% alignment improvement over the static baseline and drives diverse, h-dependent coordination decisions when evaluated on real enterprise attack sequences.

Chapter 6

Conclusion

6.1 Summary of Findings

The central research problem addressed by this thesis is the fingerprinting vulnerability of static honeypots, identified by Sun et al. [6]: because conventional honeypots present deterministic deception behaviour, adversaries can distinguish them from genuine hosts through targeted probe packets, rendering their intelligence-collection function ineffective.The secondary issue is the coordination gap in single-agent adaptive honeypot systems that, as Li et al. [2] point out, have limited computation and difficulties in responding to the multi-step, multi-vector nature of Advanced Persistent Threat (APT) campaigns.

The proposed framework addresses both problems through four integrated components: a scripted simulation environment based on the twelve-node Hawkeyes enterprise network [18], a Transformer encoder [19] that converts the attacker's live technique sequence into a 64-dimensional intent vector h, a Gated Multi-Modal Fusion Q-network that combines network telemetry and h through a learnable sigmoid gate, and an intent-conditioned QMIX mixing network [20] that coordinates the four-agent honeypot swarm under the Centralized Training, Decentralized Execution (CTDE) paradigm [10].

The experimental evaluation across five independent runs, four attacker strategies, and multiple evaluation protocols yields the following principal findings. First, both MARL methods achieve alignment scores of 0.63–0.76, compared to approximately 0.12 for the static baseline a greater than 600% improvement confirming that coordinated adaptive deception provides dramatically superior semantic fit with attacker behaviour relative to conventional static deployment. Second, the Alignment Score is identified as the only reliable discriminating metric for adaptive honeypot evaluation; DSR, Engagement Length, and Interaction Depth all overlap between MARL and static methods due to the topology of the evaluation environment. Third, COMISET fine-tuning produces a 22× perplexity reduction (133.2 to 5.9) and a Top-3 accuracy of 71.5% on held-out real enterprise data, confirming that fine-tuning is a prerequisite for the intent vector h to carry genuine predictive signal. Fourth, the framework generalises strongly to unseen attacker strategies with a 2.36% diagonal-to-off-diagonal alignment drop, below the 5% threshold for strong generalisation. Fifth, the closed-loop integration test confirms a 314% alignment improvement over the static baseline when the encoder is driven by real COMISET sequences, with the h-Variation Test passing, providing mechanistic confirmation of h-dependent swarm

coordination.

6.2 Contributions to the Field

The work presented in this thesis makes the following distinct contributions to the field of adaptive honeypot deception and multi-agent cyber defence.

The first contribution is the novel Gated Multi-Modal Fusion Q-network architecture, which combines the 310-dimensional network telemetry and the 64-dimensional Transformer intent vector h through a learnable element-wise sigmoid gate. The gate bias is initialised to $+2.0$ (yielding $\sigma(+2.0) \approx 0.880$), providing principled training stability by weighting telemetry at 88% during early training when h is uninformative, and allowing the gate to shift toward h as the encoder improves. This design corrects the gradient-suppression problem that would arise from naive 374-dimensional concatenation, where telemetry would structurally dominate 83% of the input width.

The second contribution is the extension of the QMIX mixing network [20] with h-conditioned hypernetworks. Standard QMIX conditions its hypernetworks exclusively on the global state; the proposed design concatenates h alongside the state, making the mixing weights W_1, b_1, W_2, b_2 functions of attacker intent. This enables intent-driven coordination: when h encodes credential-seeking behaviour, the mixer can upweight the Q-value of the agent best aligned with the `credential_store` action, producing intent-responsive swarm behaviour that is not achievable with state-only hypernetworks.

The third contribution is the real-data validation of the Transformer encoder through COMISET fine-tuning and closed-loop evaluation. The 22× perplexity reduction from 133.2 to 5.9, the Top-3 accuracy of 71.5% on 847 held-out sessions, and the 314% closed-loop alignment improvement collectively demonstrate that the encoder learns genuine, transferable representations of real enterprise attack behaviour — a critical validation step that distinguishes this work from simulation-only evaluations.

The fourth contribution is the empirical identification of Alignment Score as the correct primary metric for evaluating adaptive honeypot deception systems. The demonstration that DSR, Engagement Length, and Interaction Depth all produce full overlap between MARL and static methods in this topology is itself a methodological finding: it alerts future researchers to the risk of drawing incorrect conclusions from conventionally used honeypot metrics in topologies where the attacker's traversal path is topologically constrained.

The fifth contribution is the fog-of-war evaluation protocol and its findings. The demonstration that the Thesis model outperforms the NoTrans ablation in 4 of 5 independent runs at 90% telemetry obstruction, and that the RateGap (attacker confinement) is consistently negative across all runs and fog levels, establishes that the Transformer encoder provides reliable defensive value under realistic partial-observability conditions, not only in the idealised full-telemetry simulation setting.

6.3 Limitations

An honest account of the proposed framework must acknowledge four structural limitations identified through experimental evaluation.

The first limitation is the Transformer's structural deficit against random attackers. The Ran strategy produces a consistent alignment lift of -0.050 to -0.102 across all five independent runs, meaning the NoTrans ablation outperforms the Thesis model under uniformly random technique selection. This is a mathematical boundary: self-attention can only identify temporal patterns that exist in the input sequence. A uniformly random sequence contains no exploitable structure, and the encoder therefore produces an intent vector h that is informationally equivalent to noise. When this noise-h is injected into the gated fusion, it slightly degrades the policy relative to the clean $h = \mathbf{0}$ ablation.This constraint suggests a deployment boundary: the proposed framework is maximally effective against structured attack kill-chains (HiE, HiC, Mix) and should be supplemented with entropy detection or anomaly scoring for adversaries employing randomised technique selection. The second limitation is the near parity result for HiC and Mix strategies under full observability conditions. The mean Transformer lift for HiC is -0.011 and for Mix is -0.015, both near zero, suggesting that the encoder's h signal offers little advantage over $h = 0$ when telemetry is fully available. The heterogeneous within-tactic technique selection reduces the encoder's predictive precision, as demonstrated by the HiC attackers who combine lateral-movement techniques such as T1550 and T1072 with other patterns, limiting the gate's ability to shift toward h.

The third limitation is the absence of formal statistical testing. The evaluation reports ranges across five independent runs (e.g., AlignGap from -0.050 to -0.102 for Ran) rather than confidence intervals or paired t-tests. The five-run protocol offers a practical characterisation of variance but does not provide statistically rigorous significance claims. Future work should increase the number of independent runs and use appropriate hypothesis testing to confirm the directional findings.

The fourth limitation is that training and evaluation of reinforcement-learning were only performed on simulations. The scripted Hawkeyes attacker operates on fixed strategy tem- plates, whereas real APT actors are dynamically responsive to defensive responses, use custom tooling beyond the 25-token MITRE vocabulary, and operate on longer time horizons than the episode lengths used here. The reported performance metrics are thus an upper bound on the expected performance in a live deployment setting. Additional engineering effort would be required to port the framework to a live network setting, including faster inference pipelines and a security architecture for the deployed model itself.

6.4 Recommendations for Future Work

The contributions and limitations of this thesis point to four directions for future work.

The first is entropy-aware gating for random adversaries. The mechanism of the gate bias can be extended with the sequence entropy estimate calculated from the attention weights of the encoder or the prediction confidence of the next-technique head. When entropy is high (indicating a random or unpredictable sequence), the gate would automatically increase g toward 1.0 (full telemetry reliance) and ignore the noisy h signal. For low entropy (a structured kill chain), the gate would allow hto to contribute normally. Such a confidence-aware gating would remove the structural deficit observed under the Ran strategy without requiring a separate detection system.

The second direction is multi-strategy joint training with a strategy-classification auxiliary head. Rather than training a separate model for each of the four attacker strategies, a single model trained across all strategies simultaneously with an auxiliary strategy-classification loss could develop a unified policy that adapts its behaviour based on the inferred strategy. This would reduce the deployment burden (one model instead of four) and could improve generalisation by exposing the encoder to a wider distribution of technique sequences during training.
The third direction is an LSTM baseline comparison. The original design process evaluated LSTM as an alternative to the Transformer encoder but did not include a trained LSTM encoder in the quantitative results due to scope constraints. A direct comparison of the fine-tuned Transformer encoder against a fine-tuned LSTM encoder on the COMISET held-out test set would clarify the specific benefit of multi-head self-attention over recurrent architectures for kill-chain sequence modelling, complementing the qualitative arguments made in Chapter 4.
The fourth direction is live network deployment and GraphRAG integration. The simulation-only scope of this work is its most significant constraint relative to real-world applicability. Deploying the framework against real attacker tooling in a controlled environment such as a cyber range with live sensors would provide definitive evidence of its operational value. Furthermore, the GraphRAG-based multi-asset state modelling approach of SAGE [14] could be integrated to extend the honeypot swarm beyond four agents and to model complex inter-node relationships that the current flat 310-dimensional state representation cannot capture, enabling deployment at enterprise scale.

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